

DETERMINANTAL BOUNDARY ATLAS, NILPOTENT DEPTH, AND POLARIZED RECONSTRUCTION IN MULTIPLICATION FAILURE

QIAN QI

ABSTRACT. We study the nonreduced Fitting scheme on a Grassmannian on which multiplication fails for cube-zero algebras with Hilbert function $(1, 4, 6)$. The first nilpotent layer reconstructs the ramification quartic K3 surface together with its quartic polarization. Revision 127 determines substantially more of the singular Schubert boundary.

On every projection-rank chart write

$$\mathcal{I}_{\widehat{D}_R} = dJ, \quad d = \det T, \quad J = I_6[A_H, B_H(1_H \otimes T), C_H \operatorname{Sym}^2 T].$$

A determinant-completion identity gives

$$d^5 \in J$$

without a rank hypothesis. Hence the intrinsic nilradical satisfies $\widehat{\mathcal{N}}_R^6 = 0$ on the whole Grassmannian, so the normal cone has at most five positive graded layers.

At projection corank two, whenever A_H is injective, the first graded support is the unsplit quartic contact cone without any mixed-regularity assumption. The remaining boundary is organized by the rank of

$$\overline{B}_H : H \otimes (V/H) \longrightarrow S_R / \operatorname{im} A_H$$

and by the position of $\mathbb{P}(\ker \overline{B}_H)$ relative to the Segre quadric in $\mathbb{P}(H \otimes (V/H))$. This gives nine point/line/plane orbit types: nondecomposable and decomposable kernel points, secant and tangent kernel lines, the two ruling families distinguished by which tensor factor is fixed, nondegenerate and decomposable kernel hyperplanes, and the zero mixed map. On the coefficient-transverse open of each orbit type we compute the exact nilpotency index and the Hilbert function of the second and third graded supports. Rank drops of A_H are governed by a finite determinantal valuation atlas, with an explicit depth lower bound and the universal depth-five cutoff.

At projection corank three the cubic determinant power is excluded by the GL_3 -types of the leading residual minors; the local nilpotency index is therefore exactly five or six, separated by one degree-twelve Macaulay rank condition. At projection corank four the decomposition

$$\bigwedge^6 \operatorname{Sym}^2 \mathbb{C}^4 \simeq \mathbb{S}_{(5,4,2,1)} \mathbb{C}^4 \oplus \mathbb{S}_{(4,4,4,0)} \mathbb{C}^4$$

excludes d^3 . The second summand is dual, up to the determinant twist, to the Jacobian-quartic covariant of the web; its nonvanishing on G_4^0 forces $d^4 \in J$. Thus the deepest projection stratum has exact nilpotency index five.

The extended Rees algebra of the intrinsic nilradical is retained as the standard deformation-to-the-normal-cone mechanism. The manuscript-specific content is the graded boundary atlas above, the polarized K3 reconstruction, and the explicit primary and collision laws on the first singular Schubert boundary.

1. INTRODUCTION AND STATEMENTS

A degeneracy locus remembers where a map loses rank; its scheme structure can remember substantially more. In the multiplication problem studied here, the nilpotent layers of a Fitting scheme recover geometric data that disappear after reduction. For Hilbert function $(1, 4, 6)$,

Date: September 23, 2026 (Revision 127).

2020 Mathematics Subject Classification. 14M12, 14J28, 13C40, 14B05, 14D20.

Key words and phrases. Multiplication failure, Fitting scheme, nilpotent filtration, determinantal boundary, normal cone, quartic K3 surface, Schubert singularity.

the first layer recovers a polarized quartic K3 surface. The next layer detects the first singular Schubert stratum. The purpose of this revision is to determine how those two layers fit together across the rank boundary, including contact collisions and the first failure of the mixed-regular hypothesis.

Let V be a complex vector space of dimension $e \in \{3, 4\}$ and put $p = e(e-1)/2$. For an e -dimensional relation space

$$R \subset \text{Sym}^2 V$$

set

$$S_R = \text{Sym}^2 V / R, \quad B_R = \mathbb{C} \oplus V \oplus S_R,$$

where multiplication on V is the quotient map and $(V \oplus S_R)S_R = 0$. On

$$X_R = \text{Gr}(e, V \oplus S_R)$$

let $\mathcal{K}$ be the tautological subbundle and consider

$$\widehat{D}_R = V(\text{Fitt}_0 \text{ coker}[\text{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \longrightarrow B_R \otimes \mathcal{O}]), \quad m \geq 2. \quad (1.1)$$

The scheme is independent of m . Its reduction is the Schubert divisor $\widehat{\Delta}_R$ on which $\mathcal{K} \rightarrow V$ drops rank.

Regard R as a web of quadrics on $\mathbb{P}(V^*)$. On the nonempty invariant open G_e° used below the web is basepoint-free and the Jacobian hypersurface

$$Y_R = V(\det(\partial q_i / \partial x_j)) \subset \mathbb{P}(V^*) \quad (1.2)$$

is smooth. It is a genus-one curve for $e = 3$ and a quartic K3 surface for $e = 4$.

The first nilpotent layer. Let $U_R \subset X_R$ be the projection-corank-at-most-one open and write $D_R = \widehat{D}_R \cap U_R$. The local normal form on this open is

$$\mathcal{I}_{D_R} = t(t, f) = (t) \cap (t, f)^2,$$

where $t = 0$ is the Schubert divisor and $f = 0$ is the ramification equation on it. If $\mathcal{N}_R$ is the nilradical of $\mathcal{O}_{D_R}$, then

$$\mathcal{N}_R^2 = 0, \quad \text{Ann}(\mathcal{N}_R) = \mathcal{I}_{E_R} / \mathcal{I}_{D_R},$$

and E_R is a smooth intrinsic scheme.

Theorem 1.1 (Polarized reconstruction). *For $R \in G_e^\circ$, put*

$$\mathcal{L}_R = \omega_{E_R} \otimes \mathcal{N}_R^{\otimes -(p+e-1)}.$$

There is an isomorphism of graded algebras

$$\bigoplus_{n \geq 0} H^0(E_R, \mathcal{L}_R^{\otimes n}) \simeq \bigoplus_{n \geq 0} H^0(Y_R, \mathcal{O}_{Y_R}(en)).$$

Consequently D_R determines $(Y_R, \mathcal{O}_{Y_R}(e))$. For $e = 4$ it determines $(Y_R, \mathcal{O}_{Y_R}(1))$. The same conclusions hold with $\widehat{D}_R$ in place of D_R ; no isomorphism is required to preserve the ambient Grassmannian, the relation space, or any chosen projection.

The nilpotent filtration across projection rank. Let $\widehat{\mathcal{N}}_R$ denote the nilradical of $\mathcal{O}_{\widehat{D}_R}$. For $j \geq 1$ define

$$\mathcal{Z}_j(\widehat{D}_R) = V\left(\text{Ann}_{\mathcal{O}_{\widehat{D}_R}}(\widehat{\mathcal{N}}_R^j)\right). \quad (1.3)$$

These schemes are intrinsic and satisfy $\mathcal{Z}_{j+1} \subset \mathcal{Z}_j$. The global theorem below identifies the entire associated graded nilpotent algebra on the whole Grassmannian by residual colon ideals. Its subsequent clauses compute the first two nonzero pieces geometrically across the projection-corank-one/projection-corank-two boundary and through the first tensor-rank wall.

At a projection-rank-two point let $H \subset V$ be the image and $W = V/H$. Write

$$A_H : \text{Sym}^2 H \rightarrow S_R$$

for the pure H -block and

$$\overline{B}_H : H \otimes W \longrightarrow S_R / \text{im } A_H$$

for the induced mixed block. We call (R, H) *mixed-regular* when A_H is injective, $\overline{B}_H$ is surjective, and its one-dimensional kernel is generated by a rank-two tensor. This condition does not impose simple contact roots.

On a mixed-regular transverse slice write

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc, \quad \mathbf{m} = (a, b, c, d).$$

The restricted quartic $Y_R \cap \mathbb{P}(H^\perp)$ is represented by a binary quartic h_H . The root-free residual calculation defines a canonical unsplit contact ideal

$$I_H = (\delta, g_0, \dots, g_4),$$

whose projectivization on the rank-one Segre quadric is the Cartier divisor

$$V(h_H) \times \mathbb{P}^1.$$

Theorem 1.2 (Stratified nilpotent algebra at the first singular Schubert boundary). *Let $R \in G_4^\circ$.*

(1) *On the smooth locus of $(\widehat{D}_R)_{\text{red}}$,*

$$\mathcal{Z}_1(\widehat{D}_R) = E_R, \quad \widehat{\mathcal{N}}_R|_{E_R} \simeq \mathcal{O}_{E_R}(-\widehat{\Delta}_R),$$

and Theorem 1.1 reconstructs the polarized quartic K3 from this first layer.

(2) *On the whole Grassmannian set*

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{D}_R} : \mathcal{I}_{\widehat{\Delta}_R}), \quad \mathcal{L} = \mathcal{I}_{\widehat{\Delta}_R} / \mathcal{I}_{\widehat{\Delta}_R}^2,$$

and, for $j \geq 1$,

$$W_j = V_{\widehat{\Delta}_R} \left((\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1}) + \mathcal{I}_{\widehat{\Delta}_R} \right).$$

Then

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j \geq 1} \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad (1.4)$$

and

$$\text{Ann}(\widehat{\mathcal{N}}_R^j) = \frac{(\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1})}{\mathcal{I}_{\widehat{D}_R}}. \quad (1.5)$$

The sum is finite. These identities hold at every projection rank and every rank-drop point. On a Schur chart $\mathcal{I}_{\widehat{\Delta}_R} = (d)$, $\mathcal{I}_{\widehat{D}_R} = dJ$, they become

$$\text{Ann}(N^j) = (J : d^{j-1}) / (dJ), \quad N^j / N^{j+1} \simeq A / ((d) + (J : d^{j-1})),$$

and the exact nilpotency index is

$$1 + \min\{k \geq 0 : d^k \in J\}.$$

- (3) On every mixed-regular projection-rank-two chart,

$$\mathcal{I}_{\widehat{D}_R} = \delta I_H \cap \mathfrak{m}^5. \quad (1.6)$$

The factor $\mathfrak{m}^5$ is the actual irredundant $\mathfrak{m}$ -primary component throughout the collision boundary.

If

$$h_H = \prod_{\nu=1}^s \ell_\nu^{e_\nu}, \quad \sum_{\nu=1}^s e_\nu = 4,$$

the associated primes are

$$(\delta), P_1, \dots, P_s, \mathfrak{m},$$

where P_ν is the contact prime corresponding to ℓ_ν . At the generic point of P_ν , with $P_\nu = (\delta, q_\nu)$, the contact-primary type is

$$(\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1}). \quad (1.7)$$

Thus repeated contacts are controlled scheme-theoretically without choosing four distinct roots.

- (4) On the same mixed-regular neighbourhood,

$$\widehat{\mathcal{N}}_R^3 = 0 \neq \widehat{\mathcal{N}}_R^2,$$

and, writing $\mathcal{L}$ for the conormal line of the Schubert divisor and $\mathcal{C}_H = V(I_H)$ for the affine contact cone,

$$\mathrm{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{\mathcal{C}_H}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}). \quad (1.8)$$

In particular

$$\mathcal{Z}_1 = V(I_H \cap \mathfrak{m}^3), \quad \mathcal{Z}_2 = \Sigma_R$$

scheme-theoretically. On the adjacent projection-corank-one locus, $\mathcal{Z}_1$ is the ramification support E_R ; hence (1.8) describes the specialization of the first layer to the rank-two boundary and identifies the second layer there.

- (5) If $h_H \equiv 0$, equivalently the pencil $\mathbb{P}(H^\perp)$ is a line contained in Y_R , then

$$I_H = (\delta), \quad \mathcal{I}_{\widehat{D}_R} = \delta^2 \mathfrak{m} = (\delta^2) \cap \mathfrak{m}^5. \quad (1.9)$$

The containment locus in $\mathrm{Gr}(2, V)$ is canonically $F_1(Y_R)$, the finite reduced Fano scheme of lines on the smooth quartic.

- (6) Suppose A_H is injective and $\overline{B}_H$ remains surjective with one-dimensional kernel, but that kernel becomes a decomposable tensor. Under the transverse nonvanishing condition of Definition 10.4, the nilpotent index is still three while

$$\mathcal{Z}_2 = V(c, d, a^2, ab, b^2).$$

Thus the second intrinsic depth support becomes a length-three nonreduced thickening of the rank-two vertex on the first non-mixed-regular tensor-rank wall.

The global residual-colon formulas (1.4)–(1.5) are the all-corank structure theorem. Independently, the full Fitting ideal also has the block presentation of Theorem 9.1. The block presentation is used to compute the colon ideals; it is not itself substituted for their structure.

The global part of the theorem is Theorem 10.1; the geometric identification of its first pieces is proved in Sections 5–10. Thus the paper now has a canonical all-corank associated-graded object even on rank-drop loci where no closed-form primary decomposition is chosen. On the mixed-regular rank-two boundary its central identity is

$$J = I_H \cap \mathfrak{m}^3, \quad \delta J = \delta I_H \cap \mathfrak{m}^5.$$

The first equality identifies the contact saturation hidden in the root-free residual minors; the second converts it into a genuine primary statement at the rank-two vertex. It also explains why a collision alters the contact-primary type but leaves the vertex component fixed. The associated-graded formula (1.8) then follows by cancelling successive powers of δ .

The result separates three levels of information without leaving a gap in the intrinsic filtration. The residual-colon filtration and its associated graded algebra hold on the entire Grassmannian. The residual block presentation is a computational presentation of that global object. The contact-primary specialization gives an explicit geometric identification of its first two graded supports on the entire mixed-regular rank-two locus, including repeated roots and the containment locus; the decomposable-kernel theorem then computes the first tensor-rank wall beyond mixed regularity.

The contact incidence. Put $G = \mathrm{Gr}(2, V)$, with universal quotient bundle $\mathcal{Q}$. The quartic equation of Y_R restricts to a global section

$$h_R \in H^0(G, \mathrm{Sym}^4 \mathcal{Q}).$$

The incidence

$$\mathcal{R}_R = \{(H, [\alpha]) \in G \times Y_R : H \subset \ker \alpha\}$$

is a $\mathbb{P}^2$ -bundle over Y_R , hence smooth and irreducible. Over the locus where h_H is not identically zero it is finite flat of degree four over G . Its global discriminant is

$$\mathrm{Disc}(h_R) \in H^0(G, (\det \mathcal{Q})^{12}),$$

so the discriminant divisor has Pluecker class $12H$. The interpretation of the four etale sheets as the four simple-contact primary branches requires, in addition, the mixed-regular rank conditions. On the containment locus the fibre is a projective line and the finite-cover interpretation is replaced by Theorem 10.6.

Simple contacts as the split form. On the simple-contact mixed-regular open the unsplit theorem recovers the earlier etale primary decomposition.

Theorem 1.3 (Corank-two primary structure on the simple-contact open). *There is a nonempty invariant open $G_4^\dagger \subset G_4^\circ$ such that, for each $R \in G_4^\dagger$, a dense open set of rank-two images $H \subset V$ has the following property. Near every plane with image H , after an etale extension and smooth auxiliary coordinates,*

$$\mathcal{I}_{\widehat{D}_R} = (\delta) \cap P_1^2 \cap P_2^2 \cap P_3^2 \cap P_4^2 \cap \mathfrak{m}^5. \quad (1.10)$$

The rank-two stratum is an additional associated support,

$$\widehat{\mathcal{N}}_R^3 = 0 \neq \widehat{\mathcal{N}}_R^2, \quad \mathrm{Ann}(\widehat{\mathcal{N}}_R^2) = \mathfrak{m}/\mathcal{I}_{\widehat{D}_R}.$$

Theorem 1.2 is the unsplit and stratified extension: simple roots give $e_\nu = 1$ in (1.7); collisions give higher e_ν ; and $h_H \equiv 0$ gives the separate containment algebra (1.9).

Variation and finite ambiguity. The polarized K3s vary nontrivially.

Theorem 1.4 (Variation and finite ambiguity). *For $e = 4$, the image of*

$$R \longmapsto (Y_R, \mathcal{O}_{Y_R}(1))$$

in quartic K3 moduli has dimension nine. On a nonempty invariant open, the induced map from projective equivalence classes of relation spaces to its image is generically finite. Thus, over a general point of this image, D_R and $\hat{D}_R$ determine at most finitely many isomorphism classes of algebras B_R .

The exact rank-24 differential witness in Section 7 proves this assertion for the actual Jacobian quartic map. Section 14 separates that arithmetic witness from the classical geometric reason for the dimension count. A regular web gives the classical Reye Enriques surface and its K3 cover. The web parameter is 24-dimensional and the infinitesimal projective orbit is 15-dimensional, leaving the classical nine-dimensional quotient. We make the parameter-to-Hilbert bridge explicit and isolate the finite involution ambiguity on a fixed polarized K3.

The number nine is therefore not used as a novelty signal. The intrinsic statement is that an abstract nonreduced multiplication scheme recovers the polarized K3 and, in addition, the stratified nilpotent algebra described in Theorem 1.2.

Classical and priority boundaries. The bilinear incidence K3, the fixed-point-free factor-exchange involution, and the associated Reye congruence are classical. We compare them with Cossec, Arrondo–Sols, Dolgachev–Reider, Ingalls–Kuznetsov, and the modern nodal-Enriques treatment in Section 14. We likewise compare the nilpotent filtration with ribbons and multiple structures. The claim is not that nilpotent filtrations in general are new; it is that the particular Fitting algebra in (1.1) has intrinsic graded pieces that recover the polarized K3, the unsplit contact Cartier geometry, and the first singular Schubert support.

Ballico’s failure-locus work is treated as a genuine predecessor. The accessible 1996 continuation is compared at theorem level. The 1993 article is cited explicitly and is not used as a premise in any non-anticipation argument. The mathematical theorems of this paper are stated and proved independently of that documentary question.

2. THE FINITE DETERMINANTAL BOUNDARY ATLAS

The residual-colon filtration of Theorem 10.1 is intrinsic at every projection rank. The issue left open there is to replace an unbounded list of formal colons by a finite geometric atlas and to identify the first exceptional strata beyond mixed regularity. We do this in three steps. First, a determinant-completion identity gives a uniform depth bound. Second, projection corank two is organized by the projective geometry of the mixed kernel. Third, projection coranks three and four are reduced to finite determinant-weight conditions; the corank-four condition is identified with the Jacobian-quartic covariant of the web.

Keep the notation of Theorem 9.1. Thus, at a point of projection rank r , put

$$H = \text{im}(\mathcal{K} \rightarrow V), \quad W = V/H, \quad c = \dim W = 4 - r,$$

choose a Schur chart

$$M = I_H \oplus T, \quad d = \det T,$$

and write

$$\mathcal{I}_{\hat{D}_R} = dJ, \quad J = I_6 \left[A_H, B_H(1_H \otimes T), C_H \text{Sym}^2 T \right]. \quad (2.1)$$

The local nilradical is generated by the class of d .

Theorem 2.1 (Finite all-corank depth and multiplication). *For every $R \in G_4^\circ$ and every projection-rank chart,*

$$d^5 \in J. \quad (2.2)$$

Consequently

$$\widehat{\mathcal{N}}_R^6 = 0$$

on the whole Grassmannian. In particular

$$W_j = \emptyset \quad (j \geq 6),$$

and the special fibre of the intrinsic normal-cone specialization has at most five positive graded layers.

For $i, j \geq 1$, $i + j \leq 5$, the multiplication map

$$\frac{\widehat{\mathcal{N}}^i}{\widehat{\mathcal{N}}^{i+1}} \otimes \frac{\widehat{\mathcal{N}}^j}{\widehat{\mathcal{N}}^{j+1}} \longrightarrow \frac{\widehat{\mathcal{N}}^{i+j}}{\widehat{\mathcal{N}}^{i+j+1}}$$

is, under the identifications

$$\frac{\widehat{\mathcal{N}}^j}{\widehat{\mathcal{N}}^{j+1}} \simeq \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

the tensor twist of the canonical quotient multiplication

$$\mathcal{O}_{W_i} \otimes_{\widehat{\Delta}_R} \mathcal{O}_{W_j} \longrightarrow \mathcal{O}_{W_{i+j}}. \quad (2.3)$$

In particular it is surjective. Thus no additional multiplication constant is hidden after the schemes W_j are known.

We next sharpen the corank-two analysis. Assume A_H is injective. Put

$$E_H = S_R / \text{im } A_H, \quad \beta_H = \overline{B}_H : H \otimes W \longrightarrow E_H, \quad \chi_H = \overline{C}_H : \text{Sym}^2 W \longrightarrow E_H.$$

Here $\dim H = \dim W = 2$ and $\dim E_H = 3$. Let

$$Q = \mathbb{P}(H \otimes W)_{\text{rk}=1} \simeq \mathbb{P}^1 \times \mathbb{P}^1$$

be the Segre quadric.

Definition 2.2 (Mixed-kernel orbit type). The *mixed-kernel orbit type* of (R, H) is the rank $b = \text{rank } \beta_H$, together with the incidence type of $\mathbb{P}(\ker \beta_H)$ with Q .

For $b = 3$ this is a point off Q or on Q . For $b = 2$ it is a secant line, a tangent line, an H -ruling $\mathbb{P}(h_0 \otimes W)$, or a W -ruling $\mathbb{P}(H \otimes w_0)$ contained in Q . For $b = 1$, the one-dimensional image of $\beta_H^* : E_H^* \rightarrow (H \otimes W)^*$ is generated by a tensor of rank two or rank one; equivalently the kernel plane cuts Q in a smooth conic or in a pair of ruling lines. The final type is $b = 0$.

These are exactly the nine $GL(H) \times GL(W) \times GL(E_H)$ -orbits for the mixed block. The first two are the mixed-regular and decomposable-kernel cases already encountered in Section 10; the remaining seven are higher-dimensional mixed kernels or failures of mixed surjectivity.

For a homogeneous ideal $K \subset P = \mathbb{C}[a, b, c, d]$, write

$$H_{P/K}(n) = \dim_{\mathbb{C}}(P/K)_n.$$

The coefficient-transverse open in each orbit is the maximal open on which the finite Macaulay matrices in Definition 2.4 have the ranks displayed in the following theorem. The proof exhibits an integral point in each open, so none of them is empty.

Theorem 2.3 (Corank-two mixed-kernel atlas). *Assume A_H is injective. Then the first positive graded support is determined without any mixed-regularity assumption:*

$$\mathcal{O}_{W_1} \simeq P/(\delta, g_0, \dots, g_4), \quad \delta = ad - bc, \quad (2.4)$$

where, on $P/(\delta)$ with $T = uv^t$, the degree-four classes span

$$h_H(u) \operatorname{Sym}^4(\mathbb{C}_v^2).$$

Thus, if $h_H \neq 0$,

$$\operatorname{Proj} W_1 = V(h_H) \times \mathbb{P}_v^1 \subset Q,$$

whereas $h_H = 0$ gives $W_1 = V(\delta)$.

On the coefficient-transverse open of each mixed-kernel orbit, the remaining graded supports and the exact local nilpotency index are as follows.

b	mixed-kernel type	index	W_2	W_3
3	point off Q	3	length 1	empty
3	point on Q	3	length 3, $H = (1, 2)$	empty
2	secant line	3	length 8, $H = (1, 4, 3)$	empty
2	tangent line	3	length 8, $H = (1, 4, 3)$	empty
2	H -ruling $\mathbb{P}(h_0 \otimes W)$	4	length 15, $H = (1, 4, 6, 4)$	length 1
2	W -ruling $\mathbb{P}(H \otimes w_0)$	3	$H(n) = n + 1$ for $n \geq 4$	empty
1	dual tensor rank 2	4	$H(n) = 2n + 2$ for $n \geq 3$	length 1
1	dual tensor rank 1	4	$H(n) = 3n + 3$ for $n \geq 3$	length 3
0	zero mixed map	4	$V(\delta)$	$V(\delta)$

For $b = 0$, surjectivity of the original quotient multiplication forces χ_H to be an isomorphism and

$$J = (\delta^3); \quad (2.5)$$

hence the last row is an exact identity, not only a generic one.

In the H -ruling and $b \leq 1$ rows one has $h_H = 0$. On the W -ruling row the contact quartic is nonzero on the transverse open and has the root forced by w_0 . For $b = 1$, the projectivization of W_2 has degree two in the dual-rank-two case and degree three in the dual-rank-one case. The length-one and length-three third layers are respectively the reduced rank-two vertex and the length-three vertex thickening already appearing on the decomposable-kernel wall.

The complement of the coefficient-transverse open is itself part of the atlas, rather than an omitted exceptional set. We make this finite refinement explicit.

Definition 2.4 (Macaulay valuation atlas). Let $P_c = \mathbb{C}[t_{ij} : 1 \leq i, j \leq c]$, $d = \det T$, and let J be the residual ideal on a fixed projection-rank chart. For $1 \leq k \leq 5$, let

$$\mathbf{M}_k(J)$$

be the coefficient matrix whose columns are all products

$$m q, \quad q \in \mathcal{G}(J), \quad \deg m + \deg q = ck,$$

written in the monomial basis of $(P_c)_{ck}$, where $\mathcal{G}(J)$ is the displayed set of maximal minors in (2.1). Let v_k be the coefficient vector of d^k . Define

$$\varepsilon_k(J) = \operatorname{rank}[\mathbf{M}_k(J) \mid v_k] - \operatorname{rank} \mathbf{M}_k(J) \in \{0, 1\}.$$

Then

$$\varepsilon_k(J) = 0 \iff d^k \in J.$$

The locally closed loci obtained by fixing the ranks of the matrices $\mathbf{M}_k$, their augmented matrices, and the analogous matrices for the kernels

$$K_j = (J : d^{j-1}), \quad 1 \leq j \leq 5,$$

are called the *Macaulay valuation atlas*. All of their equations are minors of explicit finite coefficient matrices.

Because of Theorem 2.1, no matrix with $k > 5$ occurs. We refine these rank strata simultaneously by the flattening strata of the finite modules

$$\mathcal{M}_j = P_c / ((d) + K_j), \quad 1 \leq j \leq 5,$$

by the flattening strata of the cokernels of (2.3), and by the associated-point stratification in the following proposition.

Proposition 2.5 (Finite associated-prime refinement). *On every finite-type coefficient chart for (A_H, B_H, C_H) , the Macaulay valuation atlas admits a finite locally closed refinement with the following properties.*

- (1) *For every $j \leq 5$, $\mathcal{M}_j$ is flat over the coefficient stratum and its Hilbert polynomial is constant.*
- (2) *There are finitely many relative closed supports $Z_{j,\nu}$ such that, on every geometric fibre, the associated primes of $\mathcal{M}_j$ are exactly the generic points of the nonempty fibres $(Z_{j,\nu})_s$. After refinement, their dimensions, minimal-versus-embedded status, incidence relations, and generic lengths are constant.*
- (3) *The cokernels of all graded multiplication maps are flat over the stratum. Hence their ranks are constant.*

Consequently the radicals, minimal and embedded associated supports, multiplicities, nilpotency index, and multiplication ranks of the five possible positive graded layers have constant combinatorial type on each final atlas stratum.

This refined atlas is the canonical finite stratification used below for rank drops. It records coefficient collisions by actual specialization of the graded algebra, rather than by relabelling a generic primary decomposition.

There is also a useful degree bound that detects many strata before any Macaulay matrix is formed. Put

$$a = \text{rank } A_H, \quad q = 6 - a,$$

and, after quotienting by $\text{im } A_H$,

$$b = \text{rank } \overline{B}_H.$$

Every nonzero residual q -minor uses at most b linear mixed columns and therefore has transverse degree at least

$$\ell(a, b) = 2q - b. \tag{2.6}$$

Proposition 2.6 (Rank-drop depth bounds at corank two). *At projection corank two,*

$$\min\{k : d^k \in J\} \geq \kappa_0(a, b) := q - \left\lfloor \frac{b}{2} \right\rfloor. \tag{2.7}$$

Together with $d^5 \in J$, this leaves only the following finite possibilities when A_H drops rank:

a	q	possible $(b; k)$
2	4	(4; 2, 3, 4, 5), (3; 2; 3, 4, 5), (1; 4, 5),
1	5	(4; 3, 4, 5), (3; 2; 4, 5),
0	6	(4; 4, 5), (3; 5).

Here $k = \min\{j : d^j \in J\}$, so the nilpotency index is $k + 1$. The omitted values of b cannot occur under surjectivity of the original quadratic quotient, because the quadratic block contributes at most three independent target directions. Each comma-separated choice of k is cut out exactly by $\varepsilon_k(J) = 0$ and $\varepsilon_{k-1}(J) = 1$. In particular the extreme rank-drop type $(a, b) = (0, 3)$ has exact nilpotency index six.

At projection corank three a stronger representation-theoretic exclusion reduces the atlas to one binary choice.

Theorem 2.7 (Projection corank three). *At every projection-corank-three point of $\widehat{D}_R$, including all rank drops of A_H and of the mixed block,*

$$d^3 \notin J, \quad d^5 \in J. \quad (2.8)$$

Hence the nilpotency index is either five or six. It is five exactly when

$$\varepsilon_4(J) = 0,$$

and six exactly when $\varepsilon_4(J) = 1$. Thus the entire corank-three locus has two canonical determinantal valuation strata, with no deeper unclassified nilpotent layer.

At the deepest Schubert stratum the remaining condition is identified geometrically.

Theorem 2.8 (Projection corank four and the Jacobian quartic). *Let $R \in G_4^2$. At a projection-corank-four point, $H = 0$, $T \in \text{Mat}_{4 \times 4}$, and*

$$J = I_6(\gamma_R \text{Sym}^2 T).$$

Then

$$d^3 \notin J, \quad d^4 \in J, \quad d = \det T. \quad (2.9)$$

Consequently the local nilradical has exact index five.

More precisely, as a $GL(V)$ -module,

$$\bigwedge^6 \text{Sym}^2 V \simeq \mathbb{S}_{(5,4,2,1)} V \oplus \mathbb{S}_{(4,4,4,0)} V. \quad (2.10)$$

The absence of $(\det V)^3$ in (2.10) gives the first nonmembership in (2.9). Under the canonical duality

$$\bigwedge^6 \text{Sym}^2 V \simeq (\det V)^5 \otimes \left(\bigwedge^4 \text{Sym}^2 V \right)^*,$$

the second summand in (2.10) is dual, up to determinant twist, to

$$\det V \otimes \text{Sym}^4 V,$$

and the corresponding covariant of $\bigwedge^4 R$ is the Jacobian quartic F_R . Since $F_R \neq 0$ on G_4^2 , the determinant line $(\det V)^4$ occurs in the degree-sixteen part of J , which is exactly the second membership in (2.9).

Theorems 2.1, 2.3, 2.7, and 2.8 are the boundary classification used in this revision. The residual-colon formulas remain useful as coordinate-free identities, but the nilpotent depth is now uniformly finite, the multiplication law is explicit, the higher-dimensional mixed kernels are stratified geometrically, the A_H -rank drops have a finite determinantal valuation atlas, and the two deeper projection-corank strata have a finite exact depth classification.

3. QUADRATIC RELATIONS AND A RELATIVE RAMIFICATION FAMILY

Put $N = e(e+1)/2$ and $P = \mathbb{P}(V^*)$. Let $\mathcal{H} \subset V \otimes \mathcal{O}_P$ be the tautological hyperplane bundle. Contraction is normalized by $c_\alpha(vw) = \alpha(v)w + \alpha(w)v$.

Lemma 3.1 (Restriction and contraction). *There is a locally split exact sequence*

$$0 \longrightarrow \text{Sym}^2 \mathcal{H} \longrightarrow \text{Sym}^2 V \otimes \mathcal{O}_P \xrightarrow{c} V \otimes \mathcal{O}_P(1) \longrightarrow 0.$$

The cokernels of $\text{Sym}^2 \mathcal{H} \rightarrow S_R \otimes \mathcal{O}_P$ and $R \otimes \mathcal{O}_P \rightarrow V \otimes \mathcal{O}_P(1)$ are naturally isomorphic. Their determinant ideals define the same degree- e hypersurface when nonzero.

Proof. Locally choose ℓ with $\alpha(\ell) = 1$. The decomposition $\mathrm{Sym}^2 V = \mathrm{Sym}^2 H \oplus H\ell \oplus \mathbb{C}\ell^2$ identifies contraction with zero, the identity, and multiplication by two on the three summands, respectively. The twist accounts for rescaling α . Both cokernels are the quotient by $R + \mathrm{Sym}^2 \mathcal{H}$; hence their zeroth Fitting ideals agree. Finally, $\det c_R$ is a section of $\det V \otimes (\det R)^* \otimes \mathcal{O}_P(e)$. $\square$

Proposition 3.2. *If R is basepoint-free, its quadratic morphism $\phi_R : P \rightarrow \mathbb{P}(R^*)$ is finite and surjective of degree 2^{e-1} . Its ramification Fitting scheme is Y_R in (1.2). Moreover,*

$$R \text{ is basepoint-free} \iff \gamma_R(HV) = S_R \text{ for every hyperplane } H \subset V. \quad (3.1)$$

Proof. The pullback of $\mathcal{O}(1)$ is $\mathcal{O}_P(2)$. A positive-dimensional projective fibre contains a curve on which this ample line bundle would be trivial. Thus the proper morphism is quasi-finite, and hence finite. Its image has dimension $e - 1$ and is the whole target. Intersecting $e - 1$ pulled-back hyperplanes computes its degree.

For $J = (\partial q_i / \partial x_j)$, Euler's identity reads $J(x)x = 2(q_1(x), \dots, q_e(x))^t$. On charts where a source coordinate and a target coordinate are nonzero, choose the radial vectors first. The radial differential is a unit; its quotient is the projective differential. Removing that invertible block identifies the two ramification Fitting ideals. The contraction matrix is J^t , up to bases, so Lemma 3.1 applies.

A functional on S_R vanishing on $\gamma_R(HV)$ pulls back to one vanishing on $R + HV$. If $H = \ker \alpha$, then $(HV)^\perp = \mathbb{C}\alpha^2$. Thus such a nonzero functional exists exactly when all $q \in R$ vanish at $[\alpha]$. $\square$

Proposition 3.3 (The universal smooth open). *The set G_e° is nonempty, open, and $\mathrm{PGL}(V)$ -invariant. The determinant construction over it is a relative effective Cartier divisor $\mathcal{Y} \subset G_e^\circ \times P$, proper and smooth over G_e° . It is obtained by descent from the frame bundle of the universal quotient, and its formation commutes with arbitrary base change. Each fibre is integral, with trivial canonical bundle; for $e = 4$ it also has $H^1(\mathcal{O}) = 0$.*

Proof. Fix a p -dimensional space S and set $T = \mathrm{Hom}(\mathrm{Sym}^2 V, S)$. Restriction over $T \times P$ to $\mathrm{Sym}^2 \mathcal{H}$ is a surjective evaluation map to the rank- p^2 bundle of square matrices. The incidence of corank at least two has codimension four on each fibre over P , hence dimension $\dim T + e - 5 < \dim T$. Its projection to T is proper and has proper closed image. Similarly, HV has dimension $p + e - 1$; nonsurjectivity of its restriction has codimension e and universal dimension $\dim T - 1$. Remove these two images and nonsurjective tensors. Also require invertibility of the square restriction at one fixed hyperplane. Each is a nonempty open condition on the irreducible space T , so their intersection T_1 is nonempty.

On $T_1 \times P$ the universal determinant section is nonzero on every fibre. Since a nonzero homogeneous polynomial is a non-zero-divisor on projective space over a field, its zero scheme Z is a relative Cartier divisor and is flat over T_1 . More explicitly, apply the fibrewise flatness criterion to $0 \rightarrow \mathcal{O}(-e) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_Z \rightarrow 0$: the first two terms are base-flat and the first map stays injective on every fibre. This also proves that the same sequence stays exact after any base change.

The total space Z is smooth over $\mathbb{C}$. Indeed, evaluation is a submersion in the tensor directions, and the determinant hypersurface of square matrices is smooth at matrices of corank one; the corank-two incidence was removed. Every fibre is a nonempty degree- e hypersurface. The generic fibre over T_1 is regular, since its local rings are localizations of the regular local rings of Z . In characteristic zero it is geometrically regular, hence smooth. The nonsmooth locus of $Z \rightarrow T_1$ is closed, and its image is closed by properness and misses the generic point. Deleting that image gives a nonempty open with smooth fibres.

Surjections $\mathrm{Sym}^2 V \twoheadrightarrow S$ form the frame bundle of the rank- p universal quotient on $\mathrm{Gr}(e, \mathrm{Sym}^2 V)$. The conditions of basepoint-freeness and smooth determinant fibre depend only on the kernel. Their bad loci have closed proper projections from the corresponding projective

incidences. They therefore define the asserted open downstairs. The determinant ideal, relative Cartier property, and flatness descend along the faithfully flat frame bundle; flatness with smooth geometric fibres gives smoothness. The cokernel and Fitting constructions themselves commute with arbitrary base change. The extra fixed-hyperplane condition above is used only to produce a nonempty open in T ; it is not imposed in the definition downstairs.

A smooth positive-degree hypersurface in $\mathbb{P}^{e-1}$ is integral here: distinct positive-degree components would meet and give a singular point, and repeated components are not smooth. Adjunction gives $\omega_{Y_R} = \mathcal{O}_{Y_R}$. The hypersurface sequence gives genus one when $e = 3$ and $H^1(Y_R, \mathcal{O}) = 0$ when $e = 4$. Thus the latter is a K3 surface. Smoothness of the determinant also excludes corank at least two in the square restriction, since all cofactors then vanish. $\square$

Lemma 3.4. *The algebras B_R and $B_{R'}$ are isomorphic exactly when R and R' lie in the same $\mathrm{PGL}(V)$ -orbit.*

Proof. An isomorphism preserves the maximal ideal and its square. On their associated graded spaces it induces an isomorphism of V carrying the kernel of quadratic multiplication to the other kernel. Conversely such a linear isomorphism gives an isomorphism of the graded algebras. Scalar matrices act trivially on relation subspaces. $\square$

4. THE MULTIPLICATION IDEAL AND INTRINSIC TORSION

We first calculate on the whole Grassmannian. In this section $R \in G_e^\circ$ is fixed, and the subscript R is omitted.

Proposition 4.1 (The full Fitting presentation). *On a frame chart write $\mathcal{K} \rightarrow V \oplus S$ as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$. For every $m \geq 2$,*

$$\mathcal{I}_{\hat{D}} = (\det M)I_p(\gamma \mathrm{Sym}^2 M). \quad (4.1)$$

The reduction of $\hat{D}$ is the integral divisor $\hat{\Delta}$. Its smooth locus is the rank- $e - 1$ stratum. Consequently D is the restriction of $\hat{D}$ to the smooth locus of its reduction, and this restriction is intrinsic.

Proof. Cube-zero multiplication and the unit show, over the coefficient ring and not only on fibres, that the image of every degree $m \geq 2$ is $\mathbb{C}1 + \mathcal{K} + \mathcal{K}^2$. Its presentation is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \mathrm{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor uses the scalar column and all e columns meeting the V -rows. Expanding on these rows proves (4.1). The graph block A_0 has no effect on this ideal. Off $\det M = 0$, the quadratic block is surjective because γ is surjective, so the ideal is the unit ideal. On the divisor it vanishes. The divisor is the usual irreducible Schubert divisor: the rank- $e - 1$ stratum is dense and integral, and every lower-rank plane is a specialization of that stratum. Equivalently the standard Schubert charts give its reduced determinantal equations.

Near a point with projection rank r , an invertible r -minor reduces the projection to $\mathrm{diag}(I_r, T)$, where the entries of the $(e - r) \times (e - r)$ block T are independent transverse coordinates. One can see their independence by varying the V -components of a basis of $K \cap S$ while keeping an invertible S -frame block fixed. The determinant equation is $\det T$. Its gradient is nonzero exactly at residual corank one. This proves the smooth-locus claim, including its assertion on the Grassmannian rather than on a chosen frame space. $\square$

On Δ put $H = \text{im}(K \rightarrow V)$ and $\Lambda = K \cap S$. These give a morphism to $P \times \mathbb{P}(S)$. Define $E = \Delta \times_P Y$. The graph of K/Λ yields isomorphisms

$$\Delta \simeq \text{Tot}_{P \times \mathbb{P}(S)} \text{Hom}(\mathcal{H}, S/\Lambda), \quad (4.2)$$

$$E \simeq \text{Tot}_{Y \times \mathbb{P}(S)} \text{Hom}(\mathcal{H}, S/\Lambda). \quad (4.3)$$

Indeed the inverse construction takes the inverse image of a graph $H \rightarrow S/\Lambda$ in $H \oplus S$. Both constructions commute with base change. In particular E and Δ are smooth integral, of dimensions $ep - 2$ and $ep - 1$.

Theorem 4.2 (Ramification normal form). *On all of U the ideal of D is*

$$\mathcal{I}_D = \mathcal{I}_\Delta \mathcal{I}_E = \mathcal{I}_\Delta \cap \mathcal{I}_E^2. \quad (4.4)$$

The associated supports are exactly Δ and E . The nilradical is the invertible $\mathcal{O}_E$ -module $\mathcal{N} = \mathcal{O}_E(-\Delta)$ and is square-zero. Its annihilator recovers E as a scheme.

Proof. Near projection corank one, Gaussian operations make $M = \text{diag}(I_{e-1}, t)$. The quadratic block becomes $[A, tB, t^2c]$, with $A|_{t=0} = \gamma|_{\text{Sym}^2 H}$. Equation (3.1) says $[A, B]$ is onto locally; choose $AU + BV = I_p$. Multiplication by t then shows that $[A, tB, t^2c]$ and $[A, tI_p]$ have exactly the same column module. Their zeroth Fitting ideals agree. Off Y , A is invertible and (4.1) is (t) . On Y , the restriction has rank $p - 1$ by Proposition 3.3. A Schur complement gives a residual row with ideal (t, f) , where $f \bmod t$ is a local ramification equation. Thus the full ideal is

$$(t^2, tf) = t(t, f). \quad (4.5)$$

The pair t, f is a regular parameter pair cutting out E . The identity $(t) \cap (t, f)^2 = t(t, f)$ proves (4.4). These are primary ideals because their associated graded rings are polynomial rings over the integral smooth supports. At the generic point of E , neither primary component can be omitted. Their radicals are distinct, giving exactly the stated associated points. Finally the nilradical is $(t)/t(t, f)$; cancellation of t gives its module identification and annihilator $(t, f)/t(t, f)$. Both are intrinsic, so the calculation glues. $\square$

Lemma 4.3 (Generic torsion as a sheaf). *Let Z be noetherian with integral reduction and generic point η . For a coherent module $\mathcal{F}$ define*

$$\text{tor}_\eta(\mathcal{F}) = \ker(\mathcal{F} \longrightarrow i_* \widetilde{\mathcal{F}}_\eta), \quad i : \text{Spec } \mathcal{O}_{Z, \eta} \longrightarrow Z.$$

This kernel is coherent and is the largest coherent submodule with zero generic stalk. It is preserved by isomorphisms of schemes.

Proof. On an affine open meeting η this is the localization kernel at the prime of η . It is a submodule of a finite module over a noetherian ring, hence finite. Localization of that kernel on smaller affine opens is the corresponding kernel, so these modules glue. Every submodule with zero generic stalk lies in the kernel, and the kernel itself has zero generic stalk. The definition involves no ambient embedding or chosen primary decomposition. $\square$

Corollary 4.4 (Ordinary powers). *For $n \geq 1$, write $D^{[n]} = V(\mathcal{I}_D^n)$ and $E^{[n]} = V(\mathcal{I}_E^n)$. Then*

$$\mathcal{I}_D^n = \mathcal{I}_\Delta^n \mathcal{I}_E^n = \mathcal{I}_\Delta^n \cap \mathcal{I}_E^{2n} \quad (4.6)$$

is an irredundant primary decomposition on U . With $\mathcal{T}_n = \text{tor}_{\eta_\Delta}(\mathcal{O}_{D^{[n]}})$,

$$0 \longrightarrow \mathcal{T}_n = \mathcal{O}_{E^{[n]}}(-n\Delta) \longrightarrow \mathcal{O}_{D^{[n]}} \longrightarrow \mathcal{O}_{n\Delta} \longrightarrow 0. \quad (4.7)$$

Its annihilator defines $E^{[n]}$. The generic multiplicity on Δ is n , the nilradical index along E is $2n$, and the length of $\mathcal{T}_n$ at η_E is $n(n+1)/2$.

Proof. In the local ring let $P = (t, f)$. The associated graded ring of P is a polynomial ring in two variables over $\mathcal{O}_E$, and the initial form of t is one of those variables. Hence $(P^{2n} : t^n) = P^n$, proving $(t^n) \cap P^{2n} = t^n P^n$. The same graded argument proves that these powers are primary. The witnesses t^n and f^{2n} at η_E prove irredundancy. Localizing at (t) contracts the ideal to (t^n) , so the generic torsion is $(t^n)/t^n P^n$, and cancellation gives its annihilator P^n . The class of t has index $2n$ at E , since t^{2n} belongs to the ideal but t^{2n-1} does not. At the generic point of Δ , f is a unit. Finally $\mathcal{O}_{U, \eta_E}/(t, f)^n$ has the $n(n+1)/2$ monomials of total degree less than n as successive graded basis elements. $\square$

Lemma 4.5 (Flat flags and arbitrary base change). *Let $U \rightarrow T$ be smooth, and let $E \subset \Delta \subset U$ be relative smooth closed subschemes, each inclusion a Cartier divisor. Put $L = \mathcal{I}_\Delta$ and $P = \mathcal{I}_E$. Then $\mathcal{O}_U/P^n$ and $\mathcal{O}_U/L^n$ are flat over T . For $D_n = V(L^n P^n)$ the sequence*

$$0 \rightarrow L^n \otimes \mathcal{O}_{E^{[n]}} \rightarrow \mathcal{O}_{D_n} \rightarrow \mathcal{O}_U/L^n \rightarrow 0 \quad (4.8)$$

is exact and remains exact after any base change. Its base-changed terms are the indicated ordinary powers and products. In particular D_n is flat over T .

Proof. The immersion $E \subset U$ is regular of codimension two and has locally free conormal module. The conormal filtration of $\mathcal{O}_U/P^n$ has quotients $\text{Sym}^j(P/P^2)$ on E , for $0 \leq j < n$; these are base-flat. Induction proves flatness. The analogous principal filtration proves flatness of $\mathcal{O}_U/L^n$. Multiplication by a local equation t^n of L^n identifies the kernel in (4.8) with $L^n \otimes \mathcal{O}_{E^{[n]}}$, since t is a non-zero-divisor. Both outer terms are base-flat, so the middle term is base-flat. For a base algebra $A \rightarrow A'$, the only possible left obstruction in tensoring this short exact sequence is $\text{Tor}_1^A(\mathcal{O}_U/L^n, A')$, which is zero. The sequences defining L , P , and their powers stay exact for the same reason. Thus their images after base change are the actual ideals generated by the base-changed equations, and multiplication gives their stated products. No assertion about specialization of selected primary components is used. $\square$

Corollary 4.6. *Over G_e° , the schemes E , Δ , and U form the smooth relative flag of Lemma 4.5. All $D^{[n]}$ and $E^{[n]}$ are flat, and (4.7) commutes with arbitrary base change. Each complex fibre has (4.6).*

Proof. Use the universal quotient $\mathcal{S}$ and the graph bundles (4.2)–(4.3) over the smooth family of Proposition 3.3. The Fitting elimination works over the parameter ring, so it gives $t(t, f)$ relatively. Apply the lemma, and apply Corollary 4.4 separately on each fibre for the primary assertion. $\square$

5. THE INTRINSIC NILPOTENT-DEPTH FILTRATION

We isolate the functorial mechanism that unifies the two local models. Let Z be any noetherian scheme and let $\mathcal{N}_Z$ denote the nilradical of $\mathcal{O}_Z$. Since Z is noetherian, $\mathcal{N}_Z$ is a coherent nilpotent ideal. For $j \geq 1$ define

$$\mathcal{Z}_j(Z) = V(\text{Ann}_{\mathcal{O}_Z}(\mathcal{N}_Z^j)). \quad (5.1)$$

Lemma 5.1 (Functoriality). *Every isomorphism $f : Z \rightarrow Z'$ carries $\mathcal{Z}_j(Z)$ isomorphically to $\mathcal{Z}_j(Z')$ for every j . The construction commutes with restriction to open subsets.*

Proof. An isomorphism carries the nilradical to the nilradical, hence carries its j -th power and its annihilator ideal to the corresponding objects. Both formation of powers and annihilators commute with localization. $\square$

Apply this to $Z = \widehat{D}_R$. On the intrinsic open obtained by restricting to the smooth locus of Z_{red} , Proposition 4.1 identifies Z with D_R . Theorem 4.2 gives the local ring

$$A = \mathcal{O}/(t(t, f))$$

with nilradical generated by the class of t . Therefore

$$\text{Ann}_A((t)) = (t, f)/(t(t, f)),$$

so (5.1) recovers E_R scheme-theoretically.

Proposition 5.2 (Depth one is the polarized ramification support). *On the smooth-reduction locus of $\widehat{D}_R$,*

$$\mathcal{Z}_1(\widehat{D}_R) = E_R, \quad \widehat{\mathcal{N}}_R|_{E_R} = \mathcal{O}_{E_R}(-\widehat{\Delta}_R).$$

Consequently the pair

$$\left(\mathcal{Z}_1(\widehat{D}_R), \omega_{\mathcal{Z}_1(\widehat{D}_R)} \otimes (\widehat{\mathcal{N}}_R|_{\mathcal{Z}_1(\widehat{D}_R)})^{-9} \right)$$

is intrinsic when $e = 4$, and its graded section ring is the fourth Veronese of the polarized quartic section ring.

Proof. Only the last assertion is new notation. The annihilator calculation above is Theorem 4.2; the line-bundle identity and section-ring reconstruction are Proposition 6.2 and Theorem 1.1. $\square$

The point of (5.1) is that it makes no reference to a projection chart. The closed schemes are automatically nested:

$$\widehat{\mathcal{N}}_R^{j+1} \subset \widehat{\mathcal{N}}_R^j \implies \text{Ann}(\widehat{\mathcal{N}}_R^j) \subset \text{Ann}(\widehat{\mathcal{N}}_R^{j+1}), \quad \mathcal{Z}_{j+1} \subset \mathcal{Z}_j.$$

The new issue is therefore not formal functoriality but the scheme-theoretic computation of the layers as one crosses projection rank. Theorem 10.1 solves this globally: on the whole Grassmannian the residual colon

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{D}_R} : \mathcal{I}_{\widehat{\Delta}_R})$$

determines every annihilator and every associated-graded piece, including all projection coranks and rank-drop loci. The remaining task is geometric identification of those canonical colon schemes.

On the first higher-corank chart the next power is nonzero. The calculation in §8 gives, on the simple-contact open,

$$\widehat{\mathcal{N}}_R^3 = 0, \quad \text{Ann}(\widehat{\mathcal{N}}_R^2) = \mathcal{I}_{\Sigma_R}/\mathcal{I}_{\widehat{D}_R}.$$

Section 9 removes the simple-root assumption. Section 10 computes the two layers simultaneously across the corank-one/corank-two boundary. On the mixed-regular neighbourhood,

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{\mathcal{C}_R}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}),$$

where $\mathcal{C}_R$ is the unsplit contact cone and $\mathcal{L}$ is the Schubert conormal line. In particular $\mathcal{Z}_1$ specializes from the ramification support to its canonical contact closure with the vertex thickening forced by the Fitting algebra, while $\mathcal{Z}_2$ is the rank-two vertex. On the first tensor-rank boundary of the mixed map, $\mathcal{Z}_2$ acquires a canonical length-three nonreduced structure (Theorem 10.5).

Thus

$$\mathcal{Z}_1 \rightsquigarrow \mathcal{Z}_2$$

is not merely a comparison of two calculations on disjoint strata: it is the associated-graded specialization of one intrinsic nilpotent algebra.

Remark 5.3 (Relation with canonical filtrations on multiple structures). Nilpotent and multiple structures have long been studied through canonical filtrations and associated graded modules; ribbons are the multiplicity-two prototype, and primitive multiple schemes carry a canonical chain of lower multiplicities. See [4, 15, 11]. The construction (5.1) is deliberately more elementary: it is available for the present Fitting schemes without assuming that they are primitive, locally complete-intersection multiple structures, or Cohen–Macaulay along every higher-corank stratum. The additional content here is that consecutive annihilator layers and their associated graded

modules have a specific geometric meaning forced by multiplication: the first carries the polarized ramification K3, the second detects the rank-two Schubert support, and their specialization through contact collisions is controlled by the unsplit contact Cartier ideal rather than by a choice of primary roots.

6. RECOVERING THE POLARIZED RAMIFICATION VARIETY

Let $A = \mathcal{O}_Y(1)$ and $B = \mathcal{O}_{\mathbb{P}(S)}(1)$. In the graph description (4.3) write

$$Z = Y \times \mathbb{P}(S), \quad W = \mathcal{H}^* \otimes Q, \quad Q = (S \otimes \mathcal{O})/\Lambda, \quad \pi : E = \text{Tot}_Z W \rightarrow Z,$$

and let $\rho : E \rightarrow Y$ be the composite projection. Constant one-dimensional vector-space factors in determinant lines below are suppressed; they change none of the isomorphism classes or section rings.

Lemma 6.1 (No extra fibrewise functions). *The natural map $\mathcal{O}_Y \rightarrow \rho_* \mathcal{O}_E$ is an isomorphism.*

Proof. The assertion is local on Y , where $\mathcal{H}$ can be trivialized. If H_0 is its fibre space, the affine bundle projection gives

$$\pi_* \mathcal{O}_E = \bigoplus_{d \geq 0} \text{Sym}^d(H_0 \otimes Q^*).$$

We claim that

$$H^0(\mathbb{P}(S), \text{Sym}^d(H_0 \otimes Q^*)) = 0 \quad (d > 0). \quad (6.1)$$

The injection $Q^* \hookrightarrow S^* \otimes \mathcal{O}$ embeds a global section in the constant space $\text{Sym}^d(H_0 \otimes S^*)$. Such an element is a homogeneous polynomial on $\text{Hom}(H_0, S)$. At every line $\Lambda \subset S$, membership in $\text{Sym}^d(H_0 \otimes (S/\Lambda)^*)$ says this polynomial factors through $\text{Hom}(H_0, S/\Lambda)$; it is invariant under translations by $\text{Hom}(H_0, \Lambda)$. Taking the lines of a basis of S shows invariance under every translation of $\text{Hom}(H_0, S)$. In characteristic zero such a polynomial is constant, so its positive-degree part is zero. This proves (6.1). The degree-zero part is $\mathbb{C}$.

The same calculation tensors with the coordinate ring of each affine trivializing open of Y : cohomology on the projective space commutes with this flat extension from $\mathbb{C}$, and with the displayed direct sum in degree zero. Hence the pushforward is precisely $\mathcal{O}_Y$, with its natural multiplication. The local identifications glue. $\square$

Proposition 6.2 (The canonical nilradical line). *As line bundles on E ,*

$$\mathcal{N} \simeq \pi^*(A^{-1} \boxtimes B^{-1}), \quad \omega_E \simeq \pi^*(A^{-(p-1)} \boxtimes B^{-(p+e-1)}). \quad (6.2)$$

Consequently

$$\omega_E \otimes \mathcal{N}^{\otimes -(p+e-1)} \simeq \rho^* A^e. \quad (6.3)$$

Proof. The determinant defining $\hat{\Delta}$ is a section of $(\det \mathcal{K})^* \otimes \det V$. On E the exact sequence $0 \rightarrow \Lambda \rightarrow \mathcal{K} \rightarrow \mathcal{H} \rightarrow 0$ gives $\mathcal{I}_{\Delta}|_E = \det \mathcal{H} \otimes \Lambda$, up to a constant factor. Since $\det \mathcal{H} = A^{-1}$ and $\Lambda = B^{-1}$, Theorem 4.2 gives the first formula.

The determinant of W is $(\det \mathcal{H}^*)^{p-1} \otimes (\det Q)^{e-1} = A^{p-1} \boxtimes B^{e-1}$. The canonical bundle of a vector-bundle total space is $\pi^*(\omega_Z \otimes \det W^{-1})$, as follows by taking the determinant of its relative cotangent sequence. Here $\omega_Z = \mathcal{O}_Y \boxtimes B^{-p}$ because $\omega_Y = \mathcal{O}_Y$. This proves the second formula. Subtracting the two exponents gives (6.3): the B -exponent vanishes and the A -exponent is $-(p-1) + (p+e-1) = e$. $\square$

Proof of Theorem 1.1. An isomorphism of failure schemes preserves the nilradical and its annihilator, hence E , its invertible module $\mathcal{N}$, and its canonical bundle. It therefore preserves $\mathcal{L} = \omega_E \otimes \mathcal{N}^{-(p+e-1)}$ and its graded section ring. Proposition 6.2, Lemma 6.1, and the projection formula give

$$H^0(E, \mathcal{L}^n) = H^0(Y, A^{en} \otimes \rho_* \mathcal{O}_E) = H^0(Y, A^{en})$$

compatibly with multiplication. The full section ring of the ample line bundle A^e recovers (Y, A^e) by Proj. In this case it is also the e th Veronese of the homogeneous coordinate ring of a smooth hypersurface, so there is no ambiguity about the grading line bundle.

For a K3 surface the Picard group has no torsion. Here is the short argument needed: a nontrivial torsion line bundle T and its inverse have no nonzero sections, since a nonzero effective torsion divisor has zero intersection with an ample divisor, which is impossible. Riemann–Roch and Serre duality would then give $2 = \chi(T) = -h^1(T) \leq 0$, a contradiction. Thus an isomorphism carrying A^4 to $(A')^4$ carries A to A' . This recovers the quartic polarization, and its complete linear system recovers the quartic embedding up to projective equivalence.

Finally Proposition 4.1 recovers D from $\widehat{D}$ by restricting to the smooth locus of the reduction. This operation is intrinsic and is preserved by every abstract scheme isomorphism. The preceding proof applies to that restriction. $\square$

Remark 6.3. For $e = 3$ the theorem recovers (Y, A^3) directly. A chosen cubic root in the Picard group need not be unique. As polarized genus-one curves, however, the possible degree-three roots are isomorphic: after choosing an origin, the map $y \mapsto t_y^* A \otimes A^{-1}$ onto $\text{Pic}^0(Y)$ is multiplication by three, hence surjective. No choice of an origin is needed for the stated section-ring reconstruction. For $e = 4$ the torsion-free argument gives uniqueness without translations.

Remark 6.4 (Relation to the earlier birational argument). The compactification $\mathbb{P}_Z(\mathcal{O} \oplus W)$ is birational to $Y \times \mathbb{P}^{e(p-1)}$. Thus the earlier cancellation proof is a special case of the maximal rationally connected quotient: for a non-uniruled base, that quotient of a product with projective space is the base, birationally. See the rational-quotient construction in [14, Chapter IV, §5]. This explains the earlier abstract recovery, which is retained in the companion. The present proof does not rely on that argument: the section ring gives the actual projective model and its line bundle, not merely a birational class.

7. NINE-DIMENSIONAL VARIATION IN THE K3 CASE

We prove Theorem 1.4. Work on the standard chart of $\text{Gr}(4, \text{Sym}^2 \mathbb{C}^4)$ where the four square coefficients are the identity. Write $x = (x_0, x_1, x_2, x_3)$ and order the six cross terms as

$$m = (x_0x_1, x_0x_2, x_0x_3, x_1x_2, x_1x_3, x_2x_3).$$

For $C \in \text{Mat}_{6 \times 4}$ set

$$q_i = x_i^2 + \sum_{k=1}^6 C_{ki} m_k, \quad J_C = (\partial q_i / \partial x_j), \quad F_C = \det J_C. \quad (7.1)$$

This parametrizes the relation space, not an additional choice of its basis: the square coefficients fix that basis on this chart.

Lemma 7.1 (An integral differential witness). *At the relation system*

$$C_* = \begin{pmatrix} 2 & -1 & 2 & 0 \\ -2 & 0 & -1 & 1 \\ 2 & -1 & -2 & 0 \\ 1 & -1 & 2 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & -2 & 1 & -2 \end{pmatrix}, \quad (7.2)$$

the differential of $C \mapsto [F_C] \in \mathbb{P}(\text{Sym}^4 \mathbb{C}^4)$ has rank 24. This relation system is basepoint-free and $V(F_{C_*})$ is smooth over $\mathbb{C}$.

Proof. Let $\text{cof}_{ij}(J)$ denote the signed cofactor. The 24 parameter derivatives are

$$G_{ik} = \left. \frac{\partial F_C}{\partial C_{ki}} \right|_{C_*} = \sum_{j=0}^3 \text{cof}_{ij}(J_{C_*}) \frac{\partial m_k}{\partial x_j}. \quad (7.3)$$

Order degree-four monomials by descending lexicographic order of their exponent tuples, and form the 35×25 integral coefficient matrix D_* with columns F_{C_*} , followed by G_{ik} with $i = 0, \dots, 3$ outermost and $k = 1, \dots, 6$ innermost. The determinant of its first 25 rows and all columns is

$$4279473148893659522379284480 \neq 0. \quad (7.4)$$

Its residue modulo 101 is 41. Thus F_{C_*} and the 24 derivatives are linearly independent. Quotienting the target tangent space by the line spanned by F_{C_*} proves the differential assertion.

For completeness, Appendix B specifies the full coefficient recipe and two additional finite rank certificates. The Macaulay matrix of the q_i in degree five has rank 56, the dimension of all degree-five polynomials. The analogous matrix of the four partials of F_{C_*} in degree nine has rank 220, again the entire homogeneous piece. Explicit square minors are nonzero modulo 101, with residues 44 and 79. The integer matrices therefore have full row rank over $\mathbb{Q}$. Every monomial of those degrees belongs to the respective homogeneous ideal over $\mathbb{Q}$, and hence over $\mathbb{C}$. A common projective zero is impossible. This proves basepoint-freeness and smoothness without a numerical tolerance or a probabilistic test. $\square$

Proposition 7.2. *The Jacobian map on G_4° has a 24-dimensional image in the space of embedded smooth quartics. Its image in polarized quartic K3 moduli has dimension nine. Its general fibres after projective coordinate equivalence are finite.*

Proof. Changing a basis of R multiplies the determinant by a nonzero scalar, so the projective Jacobian map is well-defined. A change of coordinates in V acts equivariantly. Lemma 7.1 and the 24-dimensional domain show that its image has dimension 24 and that the map is generically finite onto that embedded image. The locus where the differential is injective is nonempty and open; it is invariant because the map is equivariant. The smooth admissible locus is irreducible by Proposition 3.3.

The projective stabilizer of a smooth quartic K3 is finite. Indeed, contraction with its nonzero two-form identifies its tangent bundle with its cotangent bundle, and $H^0(\Omega^1) = 0$ by $H^1(\mathcal{O}) = 0$ and Hodge symmetry over $\mathbb{C}$. The stabilizer, a closed finite-type subgroup of PGL_4 , therefore has zero-dimensional Lie algebra and is finite. Its orbits have dimension 15. Two smooth quartic embeddings define the same polarized K3 precisely when they are related by PGL_4 , because their embeddings are given by the complete polarizing linear systems. The geometric quotient of the smooth quartic locus gives the quartic K3 moduli space; equivalently one may compute on a generic geometric quotient. The invariant image consequently has dimension $24 - 15 = 9$ after quotienting. Standard background on polarized K3 moduli is given in [12, Chapter 5].

To check the finiteness assertion directly, take a general embedded quartic in the 24-dimensional image where the fibre is finite. Every relation system mapping to a projectively equivalent quartic can be moved by the corresponding projective transformation into that finite fibre. Thus there are only finitely many projective equivalence classes above the general polarized point. This argument also makes clear that the assertion is generic, rather than a statement about all exceptional fibres. $\square$

Proof of Theorem 1.4. Combine Proposition 7.2 with the recovery of the polarized quartic in Theorem 1.1 and the algebra-orbit equivalence of Lemma 3.4. $\square$

Corollary 7.3. *There is a flat family of failure schemes on the corank-one open whose canonical polarized invariant has a nine-dimensional image. In particular the associated K3 surfaces are*

not isotrivial, and the failure schemes are not all abstractly isomorphic. Their reduced divisors become the same scheme after identifying the vector spaces S_R of rank six.

Proof. Flatness is Corollary 4.6. A fixed abstract K3 cannot give a positive-dimensional algebraic family of polarization classes on that surface: its Picard scheme is discrete, since $H^1(\mathcal{O}) = 0$. Thus the positive-dimensional polarized image also implies non-isotriviality of the surfaces. Distinct values of the canonical polarized invariant give nonisomorphic failure schemes by Theorem 1.1. The graph description of Δ in (4.2) is independent of R . $\square$

8. FIXED TENSORS ACROSS PROJECTION CORANK TWO

We now work with $e = 4$ and $p = 6$. All calculations below concern a fixed quotient tensor γ_R , while its generating plane varies. They are identities of ideals, before taking radicals.

Definition 8.1. A pair (R, H) , where $R \in G_4^\circ$ and $H \subset V$ is a plane, is called admissible for the corank-two calculation if the following conditions hold. The map $A_H = \gamma_R|_{\text{Sym}^2 H}$ is injective. For a complement W to H , put $Q_H = S_R / \text{im } A_H$ and let

$$B_H : H \otimes W \longrightarrow Q_H, \quad C_H : \text{Sym}^2 W \longrightarrow Q_H$$

be induced by multiplication. The map B_H is surjective and its one-dimensional kernel is generated by a rank-two tensor in $H \otimes W$. Finally, the line of hyperplanes containing H meets Y_R in four distinct points.

The first-order map B_H is intrinsically a map from $H \otimes (V/H)$: changing a lift of a vector in V/H adds an element of $\text{Sym}^2 H$, which is zero in Q_H . Thus these conditions are independent of the chosen complement. The last condition is transverse intersection with a projective line, not smoothness of a parameter-selected polynomial without a geometric interpretation.

Lemma 8.2 (Nonemptiness of the corank-two open). *The admissible pairs form a nonempty open subset of $G_4^\circ \times \text{Gr}(2, V)$. Its projection is a nonempty invariant open $G_4^\dagger$. For each R in that image, the admissible H form a dense open subset of $\text{Gr}(2, V)$.*

Proof. Injectivity and surjectivity are open rank conditions. After taking the line kernel, tensor rank two is the nonvanishing of its determinant. On this locus the pencil restriction of the determinant is a binary quartic; nonzero discriminant is open. These conditions are invariant. Projection from a product over $\mathbb{C}$ is open, and $\text{Gr}(2, V)$ is irreducible, giving the density claim once nonemptiness is proved.

For C_* in (7.2), take $H = \langle x_0, x_1 \rangle$ and $W = \langle x_2, x_3 \rangle$. In the square-first, cross-second basis, $\gamma = (-C_* | I_6)$. The three columns for x_0^2, x_0x_1, x_1^2 have rank three. After quotienting their image, the remaining maps are

$$B_H = \begin{pmatrix} -1/2 & -1 & 1 & 0 \\ -1/2 & -1 & 0 & 1 \\ -3/2 & -2 & 0 & 0 \end{pmatrix}, \quad C_H = \begin{pmatrix} -9/2 & 0 & -1/2 \\ -5/2 & 0 & 1/2 \\ -13/2 & 1 & 7/2 \end{pmatrix}. \quad (8.1)$$

The four columns of B_H correspond to $x_0x_2, x_0x_3, x_1x_2, x_1x_3$; the three of C_H correspond to x_2^2, x_2x_3, x_3^2 . The kernel vector is $(-4, 3, 1, 1)$, whose 2×2 determinant is -7 . For $ux_2 + vx_3$, the final square column is $C_H(u^2, 2uv, v^2)^t$. The determinant of that column and the two mixed columns is

$$h(u, v) = \frac{u^4 - 8u^3v - u^2v^2 - 4uv^3 - 9v^4}{2}, \quad \text{disc } h(u, 1) = -\frac{632301}{4} \neq 0. \quad (8.2)$$

Its leading coefficient is nonzero, so there is no root at infinity. This determinant is the restriction determinant on $\text{Sym}^2(H + \mathbb{C}(ux_2 + vx_3))$ after eliminating A_H , hence defines the pencil intersection with Y_R . Lemma 7.1 places this witness in G_4° . $\square$

Lemma 8.3 (The residual matrix). *Near a plane K whose projection has admissible rank-two image H , there are smooth auxiliary parameters and four transverse coordinates a, b, c, d in which $\mathcal{I}_{\widehat{D}_R} = \delta J$, with*

$$\begin{aligned} T &= \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc, \quad \mathbf{m} = (a, b, c, d), \\ L(T) &= \begin{pmatrix} a & b & 0 & 0 \\ c & d & a & b \\ 0 & 0 & c & d \end{pmatrix}, \quad T^{(2)} = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad + bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}, \\ J &= I_3[L(T), CT^{(2)}]. \end{aligned} \tag{8.3}$$

Here C is a 3×3 matrix depending only on the auxiliary parameters, not on a, b, c, d .

Proof. Use Proposition 4.1 and an invertible projection two-minor to write $M = \text{diag}(I_2, T)$. The basis of its first image plane and a complement can be chosen independently of the Schur block T : the Gaussian row operations depend on the first two columns, and the remaining column operations remove their components from the last two columns. This proves the asserted separation of parameters.

The quadratic columns divide into $\text{Sym}^2 H$, $H \otimes W$, and $\text{Sym}^2 W$. Eliminate the injective rank-three A_H block. The residual matrix is $[B_H(I_H \otimes T), C_H T^{(2)}]$, and its maximal minors generate the residual ideal: the eliminated identity block removes a free direct summand of the cokernel. A rank-two generator of $\ker B_H$ can be put in the form $h_1 \otimes w_2 - h_2 \otimes w_1$ by invertible changes of bases. The quotient is then the symmetrization map from two two-dimensional spaces, with the displayed matrix $L(T)$. These basis changes involve only the auxiliary parameters. Finally $\det M$ is a unit times δ . This proves the ideal identity, not just its leading term. $\square$

Let

$$h(u_1, u_2) = \det \left[\begin{pmatrix} u_1 \\ u_2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ u_1 \\ u_2 \end{pmatrix}, C \begin{pmatrix} u_1^2 \\ 2u_1 u_2 \\ u_2^2 \end{pmatrix} \right]. \tag{8.4}$$

It is the binary quartic of the pencil intersection. After an étale extension of the auxiliary coefficient ring, its four roots split as distinct line subbundles $\ell_i \subset W$. Write P_i for the ideal of $\text{im } T \subset \ell_i$; it is generated by the two entries of the composite $\mathbb{C}^2 \xrightarrow{T} W \rightarrow W/\ell_i$.

Lemma 8.4 (A rank-one-cone computation). *Under the simple-root hypothesis, in the local coefficient ring of Lemma 8.3 one has*

$$I := \bigcap_{i=1}^4 P_i = (\delta, g_0, \dots, g_4), \quad J = I \cap \mathbf{m}^3 = \delta \mathbf{m} + (g_0, \dots, g_4). \tag{8.5}$$

The degree-four polynomials g_j can be chosen with restrictions $h(u_1, u_2)v_1^{4-j}v_2^j$ on $T = uv^t$.

Proof. Give a, b, c, d degree one and the auxiliary coefficients degree zero. The coordinate ring of the rank-one cone is the diagonal subring

$$A_0[a, b, c, d]/(\delta) \simeq \bigoplus_{r \geq 0} \text{Sym}^r(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v, \quad T = uv^t, \tag{8.6}$$

where A_0 denotes the auxiliary ring. This follows directly from $a = u_1 v_1$, $b = u_1 v_2$, $c = u_2 v_1$, $d = u_2 v_2$ and the single relation $ad - bc$; the monomials on either side give the same basis. In degree r , vanishing on the four distinct u -lines means divisibility by h in the u variables. The resulting module is zero for $r < 4$, and for $r \geq 4$ it is

$$h \text{Sym}^{r-4}(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v.$$

The simple-root discriminant is a unit, so the elementary divisibility and interpolation assertions used here hold over A_0 as well as over its residue field. Multiplication of binary forms shows that this ideal is generated in diagonal degree four by $hv_1^{4-j}v_2^j$. Lifting those generators proves the first identity.

The four 3-minors of the linear block $L(T)$ generate $\delta\mathfrak{m}$, by direct expansion. Modulo δ , its four columns are

$$v_1B(h_1u), \quad v_2B(h_1u), \quad v_1B(h_2u), \quad v_2B(h_2u),$$

and the three quadratic columns are $v_1^2C(u^2)$, $v_1v_2C(u^2)$, and $v_2^2C(u^2)$. Here u^2 has coefficient vector $(u_1^2, 2u_1u_2, u_2^2)^t$. A minor using one column from each of the two linear groups and one quadratic column is h times a degree-four monomial in v ; these span all five such monomials. Thus they furnish the lifts g_j . A minor with at least two quadratic columns vanishes modulo δ , and its degree is at least five, so it already lies in $\delta\mathfrak{m}$. All other minors have been accounted for. This proves $J = \delta\mathfrak{m} + (g_j)$ exactly.

Finally I is generated by its degree-two element δ and the g_j of degree four. Its intersection with $\mathfrak{m}^3$ deletes only the scalar multiple of δ in degree two, proving the second equality. Localizing in the auxiliary parameters or taking the indicated étale extension preserves these explicit identities. $\square$

Theorem 8.5 (The corank-two primary structure). *For an admissible pair the ideal δJ has the decomposition (1.10). It is irredundant. In the local failure ring its nilradical $\widehat{\mathcal{N}}$ satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathfrak{m}/(\delta J). \quad (8.7)$$

Consequently the rank-two stratum is intrinsic to this local scheme as the annihilator support of $\widehat{\mathcal{N}}^2$.

Proof. Work on a connected regular étale auxiliary chart, so that its coefficient ring is a domain. Each P_i is a prime generated by two independent linear coordinates. Its ordinary square is P_i -primary; the same argument applies to $\mathfrak{m}^5$. The ideal (δ) is prime. At the generic point of P_i , the determinant has order one, because the other row after choosing ℓ_i as a coordinate line is not zero. Therefore

$$(P_i^2 : \delta) = P_i, \quad (\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

The second equality also follows immediately from the homogeneous order two of δ and the polynomial associated graded ring of $\mathfrak{m}$. Intersecting with (δ) and cancelling δ now gives

$$(\delta) \cap \bigcap_i P_i^2 \cap \mathfrak{m}^5 = \delta \left(\bigcap_i P_i \cap \mathfrak{m}^3 \right) = \delta J$$

by Lemma 8.4.

At the generic point of the determinant divisor all the other ideals are units, so (δ) is necessary. At the generic point of P_i the other P_j and $\mathfrak{m}$ become units, and the local ideal is $\delta P_i = (\delta) \cap P_i^2$; its two components are necessary. To see that $\mathfrak{m}^5$ is necessary, note that δ^2 lies in (δ) and in every P_i^2 , but is not in δJ : $\delta \notin J$ because its degree is two. On the other hand $\mathfrak{m}\delta^2 \subset \delta J$ since $\delta\mathfrak{m} \subset J$. This also proves that the annihilator of the class of δ^2 is exactly $\mathfrak{m}$; a nonzero auxiliary coefficient cannot kill that degree-four class.

The radical of the failure ideal is (δ) , so the nilradical is generated by the class of δ . Its square is the nonzero class just considered and its cube vanishes, since $\delta^2 \in \delta\mathfrak{m} \subset J$. This proves (8.7). Over the auxiliary ring the square is free of rank one on the rank-two stratum. Intrinsically its transition functions are those of $\mathcal{L}_{\Delta}^2$ restricted to that stratum; thus it is $\mathcal{O}_{\Sigma}(-2\widehat{\Delta})$ on the admissible open Σ . The generic transverse length is one. $\square$

Proof of Theorem 1.3. Lemma 8.2 supplies the asserted nonempty open class. Lemmas 8.3 and 8.4 and Theorem 8.5 supply the fixed-tensor calculation on every indicated frame chart. The

graph coordinates are smooth auxiliary variables, so no transverse component has been discarded. The individual P_i are allowed to permute under monodromy; the intersection and the intrinsic annihilator in (8.7) descend. Four local branches need not be four distinct global irreducible components. This is why the primary decomposition was stated étale locally rather than with globally labelled roots. $\square$

9. RESIDUAL PRESENTATIONS AND THE RANK-TWO CONTACT BOUNDARY

We first record the projection-rank block presentation inherited from the full Fitting formula, and then isolate the rank-two structure to which it gives rise. The distinction between a presentation theorem and a structure theorem is important: the block formula below is exact at every projection rank, whereas the associated-prime, collision, and nilpotent-filtration theorems are proved on the loci explicitly stated. Section 10 pushes the structure theorem through all mixed-regular contact collisions, through the containment locus, and across the first non-mixed-regular tensor-rank wall.

For $0 \leq r \leq 3$, let $\Sigma_R^{(r)} \subset X_R$ be the locally closed stratum on which the projection $\mathcal{K} \rightarrow V$ has rank exactly r . If $K \in \Sigma_R^{(r)}$ has image $H \subset V$, choose a complement $V = H \oplus W$, where $c = \dim W = 4 - r$. On a frame chart around K , an invertible r -minor and Schur reduction put the projection block in the form

$$M = \begin{pmatrix} I_H & 0 \\ 0 & T \end{pmatrix}, \quad T \in \text{Mat}_{c \times c}.$$

Decompose the quotient multiplication tensor according to

$$\text{Sym}^2 V = \text{Sym}^2 H \oplus (H \otimes W) \oplus \text{Sym}^2 W$$

and write the corresponding blocks as A_H, B_H, C_H .

Theorem 9.1 (Projection-rank residual block presentation). *For every $R \in G_4^2$, every $0 \leq r \leq 3$, and every $K \in \Sigma_R^{(r)}$, one has on the preceding frame chart, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\det T) I_6[A_H, B_H(1_H \otimes T), C_H \text{Sym}^2 T]. \quad (9.1)$$

The formula is independent of any injectivity, mixed-rank, tensor-rank, or contact-transversality assumption. Thus every projection-rank stratum has the same exact block presentation. No claim of a primary or associated-prime classification on the non-mixed-regular corank-three or corank-four boundary is made by this formula alone; the separate structure theorems below derive additional information where the residual algebra can be determined.

Proof. Proposition 4.1 gives on every frame chart

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6(\gamma_R \text{Sym}^2 M).$$

The Schur operations placing M in the displayed block form are invertible and therefore change neither the maximal-minor ideal nor the local scheme. On $\text{Sym}^2 H \oplus (H \otimes W) \oplus \text{Sym}^2 W$ the induced map $\text{Sym}^2 M$ is respectively

$$I_{\text{Sym}^2 H}, \quad 1_H \otimes T, \quad \text{Sym}^2 T.$$

Since $\det M$ is a unit times $\det T$, substitution proves (9.1). $\square$

We now specialize this exact all-corank formula to $r = 2$. Write $\Sigma_R = \Sigma_R^{(2)}$. Then $c = 2$, and after choosing bases

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = \det T = ad - bc.$$

Decompose the quotient multiplication tensor according to

$$\mathrm{Sym}^2 V = \mathrm{Sym}^2 H \oplus (H \otimes W) \oplus \mathrm{Sym}^2 W$$

and write

$$A_H : \mathrm{Sym}^2 H \rightarrow S_R, \quad B_H : H \otimes W \rightarrow S_R, \quad C_H : \mathrm{Sym}^2 W \rightarrow S_R.$$

Theorem 9.2 (Exact presentation on every rank-two chart). *In the preceding coordinates one has, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\delta) I_6[A_H, B_H(1_H \otimes T), C_H \mathrm{Sym}^2 T]. \quad (9.2)$$

The formula is valid at every projection-rank-two point; no injectivity, mixed-rank, tensor-rank, or simple-contact hypothesis is required.

Proof. Proposition 4.1 gives

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6(\gamma_R \mathrm{Sym}^2 M).$$

For $M = I_H \oplus T$, the induced map on the three summands of $\mathrm{Sym}^2(H \oplus W)$ is respectively

$$I_{\mathrm{Sym}^2 H}, \quad 1_H \otimes T, \quad \mathrm{Sym}^2 T.$$

Substitution gives (9.2). All row and column operations used to reach the block form are invertible, so the maximal-minor ideal is unchanged. $\square$

Thus the exceptional rank-two locus is included in the exact presentation. Determining radicals, associated primes, and nilpotent layers of its residual factor is a separate problem. The mixed-regular case and its first tensor-rank boundary are solved below and in Section 10.

Definition 9.3. A pair (R, H) is *mixed-regular* if A_H is injective and, after quotienting S_R by its image, the induced map

$$\overline{B}_H : H \otimes (V/H) \longrightarrow S_R / \mathrm{im} A_H$$

is surjective with one-dimensional kernel generated by a rank-two tensor. No hypothesis is imposed on the multiplicities of the four ramification contacts.

These are precisely the first three conditions in Definition 8.1; the discriminant condition has been removed. On this locus the same Gaussian elimination as in Lemma 8.3 is valid and yields

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = I_3[L(T), C T^{(2)}],$$

with $L(T)$ and $T^{(2)}$ as in (8.3).

Lemma 9.4 (Root-free residual normal form). *Over the local auxiliary coefficient ring of a mixed-regular pair,*

$$J = \delta \mathbf{m} + (g_0, \dots, g_4), \quad \mathbf{m} = (a, b, c, d), \quad (9.3)$$

where each g_i is homogeneous of transverse degree four. Modulo (δ) , under the Segre parametrization $T = uv^t$, their classes span

$$h_H(u) \cdot \mathrm{Sym}^4(\mathbb{C}_v^2).$$

Here h_H is the binary quartic obtained by restricting the ramification equation to the pencil of hyperplanes containing H . The identity is valid with repeated roots, and also when the discriminant of h_H vanishes.

Proof. The four 3-minors of the linear block $L(T)$ generate $\delta\mathfrak{m}$, exactly as in Lemma 8.4. Modulo δ , write $T = uv^t$. The two pairs of linear columns are scalar multiples, by v_1, v_2 , of the two mixed columns determined by u , while the three quadratic columns are scalar multiples, by v_1^2, v_1v_2, v_2^2 , of $C(u^2)$. A minor containing one column from each linear pair and one quadratic column is therefore $h_H(u)$ times a degree-four monomial in v ; the five choices span $\text{Sym}^4(\mathbb{C}_v^2)$.

Every minor containing at least two quadratic columns vanishes on the rank-one cone, hence is divisible by the prime equation δ . Its transverse degree is at least five, so the quotient by δ lies in $\mathfrak{m}$. The remaining minors are the already listed linear and mixed ones. This proves (9.3). No factorization of h_H was used. When its roots are distinct, factorizing h_H recovers Lemma 8.4. $\square$

Theorem 9.5 (Depth two across contact collisions). *At every mixed-regular projection-rank-two point, the radical of $\mathcal{I}_{\widehat{D}_R}$ is (δ) and the nilradical $\widehat{\mathcal{N}}$ of the local failure ring satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathfrak{m}/(\delta J).$$

Consequently $\mathcal{Z}_2(\widehat{D}_R)$ is the rank-two stratum on the mixed-regular locus, even when the contact quartic has multiple roots. Moreover

$$\widehat{\mathcal{N}}^2|_{\Sigma_R} \simeq \mathcal{I}_{\Delta_R}^2|_{\Sigma_R}.$$

Proof. By Lemma 9.4, $J \subset \mathfrak{m}^3$ and $\delta\mathfrak{m} \subset J$. Since $\delta \in \mathfrak{m}^2$, we have $\delta^2 \in J$, hence $\delta^3 \in \delta J$. On the other hand $\delta \notin J$ by transverse degree, so $\delta^2 \notin \delta J$. Since $\delta J \subset (\delta)$ and a power of δ belongs to δJ , the radical is exactly (δ) . Thus the nilradical is generated by the class of δ and has exact index three.

For the annihilator, cancellation in the regular local coefficient ring gives

$$(\delta J : \delta^2) = (J : \delta).$$

The inclusion $\mathfrak{m} \subset (J : \delta)$ follows from $\delta\mathfrak{m} \subset J$. Conversely, use the transverse grading: J has no component of transverse degree two, whereas δ has degree two. If the transverse degree-zero part of r were nonzero, the degree-two part of $r\delta$ could not lie in J . Hence $r \in \mathfrak{m}$, proving $(J : \delta) = \mathfrak{m}$.

Finally δ is a local equation of the Schubert divisor $\widehat{\Delta}_R$. Therefore the generator δ^2 of the square of the nilradical transforms as the square of the conormal line. This gives the displayed line bundle after restriction to the rank-two stratum. $\square$

We next globalize the four contacts without labelling them. Put

$$G = \text{Gr}(2, V)$$

and let $\mathcal{Q}$ be its universal rank-two quotient bundle. The quartic equation F_R of $Y_R \subset \mathbb{P}(V^*)$ restricts to every line of hyperplanes containing H . Equivalently it defines a global binary quartic

$$h_R \in H^0(G, \text{Sym}^4 \mathcal{Q}). \quad (9.4)$$

Define the contact incidence

$$\mathcal{R}_R = \{(H, [\alpha]) \in G \times Y_R : H \subset \ker \alpha\}. \quad (9.5)$$

Proposition 9.6 (The global contact cover). *The projection $\mathcal{R}_R \rightarrow Y_R$ is a $\mathbb{P}^2$ -bundle; in particular $\mathcal{R}_R$ is smooth and irreducible. Over the open set $G_R^{\text{fin}} \subset G$ on which the binary quartic (9.4) is not identically zero, the map*

$$\mathcal{R}_R \longrightarrow G$$

is finite flat of degree four. There is a globally defined discriminant divisor

$$\mathfrak{D}_R = V(\text{Disc}(h_R)), \quad \text{Disc}(h_R) \in H^0(G, (\det \mathcal{Q})^{12}).$$

Whenever this section is not identically zero, $[\mathfrak{D}_R] = 12H$ in the Pluecker class group. On the finite-flat locus G_R^{fin} , the support of the branch divisor is the restriction of $\mathfrak{D}_R$; away from $\mathfrak{D}_R$ the degree-four cover is finite etale. The monodromy action on the four sheets over that etale locus is transitive.

Proof. For $[\alpha] \in Y_R$, the fibre of $\mathcal{R}_R \rightarrow Y_R$ consists of the two-planes in the three-dimensional hyperplane $\ker \alpha$, hence is $\text{Gr}(2, \ker \alpha) \simeq \mathbb{P}^2$. These fibres form the relative Grassmannian of the rank-three hyperplane bundle on Y_R , proving smoothness and irreducibility.

For fixed H , the other projection has fibre $Y_R \cap \mathbb{P}(H^\perp)$, the zero scheme of the restricted quartic. On the locus where the restriction is nonzero this is a relative effective Cartier divisor of degree four in the $\mathbb{P}^1$ -bundle $\mathbb{P}(\mathcal{Q}^*)$, hence finite flat of degree four.

The discriminant of a binary quartic is homogeneous of degree six in its coefficients and is a relative invariant of determinant weight twelve. Applied to the section of $\text{Sym}^4 \mathcal{Q}$ in (9.4), it is therefore a section of $(\det \mathcal{Q})^{12}$. Since $\det \mathcal{Q}$ is the Pluecker line bundle up to the constant factor $\det V$, its zero divisor has class $12H$ whenever it is nonzero. On G_R^{fin} the fibre is a degree-four Cartier divisor on a projective line, so its non-etale locus is exactly the multiple-root locus, i.e. the restriction of the discriminant locus. The total incidence is irreducible; therefore the monodromy action of the etale restriction on a geometric fibre is transitive. $\square$

For R in the nonempty open $G_4^\dagger$ of Lemma 8.2, the discriminant in Proposition 9.6 is not identically zero. Let $G_R^{\text{mr}} \subset G$ denote the mixed-regular locus of Definition 9.3. The contact sheets themselves split etale locally over $G_R^{\text{fin}} \setminus \mathfrak{D}_R$. Their interpretation as the four primary ideals P_i of Theorem 8.5, however, also requires the independent mixed-regular rank conditions. Thus the exact locus for that primary decomposition is

$$G_R^{\text{mr}} \cap (G_R^{\text{fin}} \setminus \mathfrak{D}_R),$$

not the discriminant complement by itself. On this intersection the P_i are the four etale contact sheets and their permutation is the monodromy of the canonical cover (9.5).

The collision divisor has the five nonzero fibre types corresponding to the partitions

$$1 + 1 + 1 + 1, \quad 2 + 1 + 1, \quad 2 + 2, \quad 3 + 1, \quad 4$$

of the degree of a binary quartic, together with the separate containment locus $h_H \equiv 0$. Theorem 9.5 does not change across these collision types. What changes is the primary splitting of the contact part, so we do not force the simple-root intersection formula onto the boundary.

The ordinary fifth power of the rank-two Schubert ideal is of course globally defined. The stronger assertion that it remains the actual $\mathfrak{m}$ -primary component of the failure ideal through every mixed-regular collision, including the zero-quartic containment locus, is proved in Theorem 10.2; it is not inferred merely from the existence of the global ideal. The same theorem gives the exact collided primary types and separates the finite contact geometry from the containment geometry.

Corollary 9.7 (Rank-two part of Theorem 1.2). *The contact-incidence and collision assertions of Theorem 1.2 hold on the stated mixed-regular locus.*

Proof. Theorem 9.5 identifies depth two and its line bundle, while Proposition 9.6 globalizes the contact incidence and discriminant. The global residual-colon filtration, the simultaneous identification of the first two graded pieces, the collided primary components, the containment locus, and the decomposable-kernel wall are proved in Section 10. $\square$

10. THE STRATIFIED NILPOTENT ALGEBRA AT PROJECTION CORANK TWO

The root-free formula of Lemma 9.4 contains more scheme structure than the nilpotency-index statement extracted from it in the preceding revision. We record that structure before passing to the exceptional mixed-kernel boundary.

Theorem 10.1 (Global residual-colon filtration). *Let $X = X_R$, let $\widehat{\Delta}_R \subset X$ be the reduced Schubert divisor, and put*

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{\Delta}_R} : \mathcal{I}_{\widehat{\Delta}_R}) \subset \mathcal{O}_X.$$

Then

$$\mathcal{I}_{\widehat{\Delta}_R} = \mathcal{I}_{\widehat{\Delta}_R} \mathcal{J}_R^{\text{res}}. \quad (10.1)$$

For every $j \geq 1$,

$$\text{Ann}_{\mathcal{O}_{\widehat{\Delta}_R}}(\widehat{\mathcal{N}}_R^j) = \frac{(\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1})}{\mathcal{I}_{\widehat{\Delta}_R}}. \quad (10.2)$$

Define the closed subscheme of the reduced divisor

$$W_j = V_{\widehat{\Delta}_R} \left((\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1}) + \mathcal{I}_{\widehat{\Delta}_R} \right). \quad (10.3)$$

If

$$\mathcal{L} = \mathcal{I}_{\widehat{\Delta}_R} / \mathcal{I}_{\widehat{\Delta}_R}^2$$

is the conormal line of the Schubert divisor, then there are canonical isomorphisms

$$\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1} \simeq \mathcal{L}^{\otimes j} \otimes_{\mathcal{O}_{\widehat{\Delta}_R}} \mathcal{O}_{W_j}. \quad (10.4)$$

Consequently

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{\Delta}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j \geq 1} \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad (10.5)$$

with the multiplication induced by multiplication of powers of $\mathcal{I}_{\widehat{\Delta}_R}$. The sum is finite.

These formulas hold on the whole Grassmannian, including every projection corank and every rank-drop locus. On a Schur chart with local equation $d = \det T$ and residual ideal J , they become

$$\text{Ann}(N^j) = (J : d^{j-1}) / (dJ), \quad N^j / N^{j+1} \simeq A / ((d) + (J : d^{j-1})). \quad (10.6)$$

In particular the nilpotency index at the point is

$$1 + \min\{k \geq 0 : d^k \in J\}. \quad (10.7)$$

Proof. Proposition 4.1 shows locally that $\mathcal{I}_{\widehat{\Delta}_R} = dJ$, where d is a local equation of the Cartier divisor $\widehat{\Delta}_R$. Since d is a non-zero-divisor in the regular local ring of X ,

$$(dJ : d) = J.$$

Thus J is the local expression of the intrinsic colon $\mathcal{J}_R^{\text{res}}$, and the local identities glue to (10.1).

In $A/(dJ)$ the nilradical is the image of (d) , because the reduction is $A/(d)$. Therefore

$$\text{Ann}(N^j) = (dJ : d^j) / (dJ) = (J : d^{j-1}) / (dJ),$$

which proves (10.2). For the graded piece, the map

$$A \longrightarrow N^j / N^{j+1}, \quad r \longmapsto rd^j$$

has kernel

$$(d) + (J : d^{j-1}).$$

Indeed

$$rd^j \in d^{j+1} + dJ \iff rd^{j-1} \in d^j + J,$$

and the latter is equivalent to writing $r = ds + t$ with $td^{j-1} \in J$. This proves the local form of (10.4). On overlaps local equations of the Cartier divisor differ by units, and d^j transforms by the j -th tensor power of the conormal transition function. Hence the local identifications glue to (10.4) and (10.5). Nilpotence of the nilradical makes the sum finite, and the same cancellation gives (10.7). $\square$

The theorem converts the globally defined annihilator filtration into an exact residual-colon filtration. What remains is to identify the colon schemes W_j geometrically on the successive rank strata. For projection corank one, $W_1 = E_R$ and all higher W_j are empty. The next two results compute W_1, W_2 through the entire mixed-regular rank-two boundary and then across its first tensor-rank wall.

Work on a mixed-regular rank-two chart. Let A_0 be the regular auxiliary coefficient ring, put

$$P = A_0[a, b, c, d], \quad \mathfrak{m} = (a, b, c, d), \quad \delta = ad - bc,$$

and localize, or complete, at the point under consideration. Thus

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = \delta \mathfrak{m} + (g_0, \dots, g_4),$$

where the g_i have transverse degree four. Define the *unsplit contact ideal*

$$I_H = (\delta, g_0, \dots, g_4). \quad (10.8)$$

Modulo δ , the Segre parametrization $T = uv^t$ identifies the homogeneous coordinate ring with

$$P/(\delta) \simeq \bigoplus_{n \geq 0} \text{Sym}^n(\mathbb{C}_u^2) \otimes \text{Sym}^n(\mathbb{C}_v^2),$$

and the image of I_H in degree n is zero for $n < 4$ and

$$h_H \text{Sym}^{n-4}(\mathbb{C}_u^2) \otimes \text{Sym}^n(\mathbb{C}_v^2) \quad (n \geq 4). \quad (10.9)$$

Theorem 10.2 (Saturated contact ideal and the fixed vertex component). *At every mixed-regular rank-two point one has the exact identities*

$$J = I_H \cap \mathfrak{m}^3 \quad (10.10)$$

and

$$\mathcal{I}_{\widehat{D}_R} = \delta I_H \cap \mathfrak{m}^5. \quad (10.11)$$

If $h_H \not\equiv 0$, the ideal I_H is $\mathfrak{m}$ -saturated. Put $\mathcal{C}_H = V(I_H)$ for the affine contact cone. Its projectivization on the Segre quadric

$$Q = \mathbb{P}_u^1 \times \mathbb{P}_v^1 \subset \mathbb{P}^3$$

is the effective Cartier divisor

$$C_H^{\text{proj}} = V(h_H) \times \mathbb{P}_v^1 \quad (10.12)$$

of class $(4, 0)$. Consequently δI_H is also $\mathfrak{m}$ -saturated, and $\mathfrak{m}^5$ in (10.11) is an actual irredundant $\mathfrak{m}$ -primary component, not merely a globally defined ideal.

If

$$h_H = \prod_{\nu=1}^s \ell_\nu^{e_\nu}, \quad \sum_{\nu} e_\nu = 4,$$

over an etale coefficient extension, and P_ν denotes the height-two contact prime corresponding to ℓ_ν , then

$$\text{Ass}(P/\mathcal{I}_{\widehat{D}_R}) = \{(\delta), P_1, \dots, P_s, \mathfrak{m}\}. \quad (10.13)$$

At the generic point of P_ν , choose a second regular parameter q_ν so that $P_\nu = (\delta, q_\nu)$. The P_ν -primary part of δI_H has the exact two-parameter normal form

$$Q_{\nu, e_\nu} = (\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1}). \quad (10.14)$$

For $e_\nu = 1$ this is P_ν^2 ; for a collided contact it is the corresponding non-ordinary primary thickening.

If $h_H \equiv 0$, then

$$I_H = (\delta), \quad J = \delta \mathfrak{m}, \quad \mathcal{I}_{\widehat{D}_R} = \delta^2 \mathfrak{m} = (\delta^2) \cap \mathfrak{m}^5. \quad (10.15)$$

In particular the rank-two vertex component remains $\mathfrak{m}^5$ on the containment locus.

Proof. The equality (10.10) is a graded calculation. The ideal I_H is generated in transverse degrees two and four, its only degree-two generator being δ . Intersecting with $\mathfrak{m}^3$ therefore removes exactly the scalar multiple of δ and leaves $\delta\mathfrak{m} + (g_0, \dots, g_4) = J$.

Since the initial form of δ has degree two and is nonzero in the polynomial associated graded ring,

$$(\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

Cancellation in the regular ring gives

$$\delta I_H \cap \mathfrak{m}^5 = \delta(I_H \cap (\mathfrak{m}^5 : \delta)) = \delta(I_H \cap \mathfrak{m}^3) = \delta J,$$

which proves (10.11).

Assume first $h_H \neq 0$. Formula (10.9) is precisely the section ideal of $\mathcal{O}_Q(-4, 0) \subset \mathcal{O}_Q$, cut out by the nonzero section h_H of $\mathcal{O}_Q(4, 0)$. Hence it is saturated and defines the Cartier divisor (10.12). The inverse image $I_H \subset P$ is therefore $\mathfrak{m}$ -saturated. The product δI_H is saturated as well: if $\mathfrak{m}^N f \subset \delta I_H$, choose a coordinate not divisible by the prime element δ to see that $\delta \mid f$; after cancelling δ , saturation of I_H applies. Thus $\mathfrak{m} \notin \text{Ass}(P/\delta I_H)$. On the other hand $\delta^2 \in \delta I_H$ has transverse degree four and does not belong to $\mathfrak{m}^5$, so the $\mathfrak{m}^5$ factor in (10.11) is irredundant.

The divisor C_H^{proj} has components $e_\nu(\{\ell_\nu = 0\} \times \mathbb{P}^1)$. Hence P/I_H is Cohen–Macaulay of pure codimension two and its associated primes are exactly the P_ν . Multiplication by δ gives an exact sequence

$$0 \longrightarrow P/I_H \xrightarrow{\delta} P/\delta I_H \longrightarrow P/(\delta) \longrightarrow 0.$$

It follows, and localization shows equality, that the associated primes of $P/\delta I_H$ are $(\delta), P_1, \dots, P_s$. Intersecting with the irredundant $\mathfrak{m}$ -primary ideal $\mathfrak{m}^5$ gives (10.13).

At the generic point of P_ν , the Cartier divisor has equation $q_\nu^{e_\nu}$, so

$$(I_H)_{P_\nu} = (\delta, q_\nu^{e_\nu}), \quad (\delta I_H)_{P_\nu} = (\delta^2, \delta q_\nu^{e_\nu}).$$

The ideal

$$(\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1})$$

is P_ν -primary, and its colon by δ is $(\delta, q_\nu^{e_\nu})$. Therefore

$$(\delta) \cap Q_{\nu, e_\nu} = \delta(\delta, q_\nu^{e_\nu}),$$

which proves the asserted generic primary normal form.

If $h_H = 0$, every g_i vanishes modulo δ . Since it has transverse degree four, $g_i = \delta q_i$ with $q_i \in \mathfrak{m}^2$. Thus $I_H = (\delta)$ and $J = \delta\mathfrak{m}$. Finally

$$(\delta^2) \cap \mathfrak{m}^5 = \delta^2(\mathfrak{m}^5 : \delta^2) = \delta^2\mathfrak{m},$$

again by the polynomial associated graded ring. This is (10.15). $\square$

The preceding theorem gives the requested specialization law without labelling roots. A collision changes the Cartier multiplicities in C_H and hence the primary type (10.14); it does not change the vertex component. The zero quartic is the unique point at which the Cartier contact divisor ceases to be a divisor and fills the whole Segre quadric.

Theorem 10.3 (Associated graded nilpotent algebra across coranks one and two). *On a mixed-regular rank-two chart let*

$$A = P/(\delta J), \quad N = \sqrt{0_A}.$$

Then $N = (\delta)/(\delta J)$, $N^3 = 0 \neq N^2$, and

$$N/N^2 \simeq P/I_H, \tag{10.16}$$

$$N^2 \simeq P/\mathfrak{m}. \tag{10.17}$$

Equivalently, with $\mathcal{L}$ the conormal line of the Schubert divisor, the associated graded algebra has the intrinsic form

$$\mathrm{gr}_N \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{\mathcal{C}_H}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}) \quad (10.18)$$

on the mixed-regular neighbourhood, with multiplication induced by the incidence of the affine contact cone $\mathcal{C}_H$ with the rank-two vertex Σ_R . Moreover

$$\mathrm{Ann}_A(N) = J/(\delta J), \quad \mathrm{Ann}_A(N^2) = \mathfrak{m}/(\delta J), \quad (10.19)$$

so

$$\mathcal{Z}_1 = V(I_H \cap \mathfrak{m}^3), \quad \mathcal{Z}_2 = V(\mathfrak{m}).$$

On the punctured determinant cone, i.e. on the adjacent projection-corank-one locus, $\mathcal{Z}_1$ is exactly the ramification support E_R . Thus $\mathcal{C}_H$ is its scheme-theoretic closure away from the vertex, while $\mathcal{Z}_1$ records in addition the canonical third-order vertex thickening forced by the Fitting ideal.

Proof. The radical of δJ is (δ) , so the nilradical is generated by the class of δ . The isomorphism (10.16) follows by cancelling one factor:

$$N/N^2 = \frac{(\delta)}{(\delta^2) + (\delta J)} \simeq \frac{P}{(\delta) + J} = \frac{P}{I_H}.$$

For the square, multiplication by δ^2 identifies its kernel with

$$(\delta J : \delta^2) = (J : \delta) = \mathfrak{m}$$

by Theorem 9.5; hence (10.17). The transition functions of the local generator δ are those of the conormal line, giving (10.18).

Cancellation also gives

$$(\delta J : \delta) = J,$$

while the preceding colon calculation gives the second formula in (10.19). Away from the vertex $\mathfrak{m}$ becomes the unit ideal, so $J = I_H$. On the smooth rank-one part of the determinant divisor, the corank-one normal form $t(t, f)$ identifies this contact equation with the ramification equation. Hence the restriction is precisely E_R . $\square$

We finally cross the first non-mixed-regular wall. This is the rank-one tensor wall in the one-dimensional mixed kernel, explicitly requested by the referee.

Definition 10.4. Assume A_H is injective and $\overline{B}_H$ is surjective with one-dimensional kernel generated by a decomposable tensor $h_1 \otimes w_1$. Choose h_2, w_2 completing bases and put

$$q_1 = \overline{B}_H(h_1 \otimes w_2), \quad q_2 = \overline{B}_H(h_2 \otimes w_1).$$

We call the point *transverse decomposable-kernel regular* if

$$\det[q_1, q_2, \overline{C}_H(w_1^2)] \neq 0.$$

This condition is independent of the chosen completing vectors. It says that the contact root forced by the decomposable kernel is simple in the direction transverse to the kernel wall.

Theorem 10.5 (The decomposable mixed-kernel wall). *At a transverse decomposable-kernel regular rank-two point there are bases for which*

$$L_{\mathrm{dec}}(T) = \begin{pmatrix} c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{pmatrix}. \quad (10.20)$$

Let

$$J_{\mathrm{dec}} = I_3[L_{\mathrm{dec}}(T), C T^{(2)}], \quad K_{\mathrm{dec}} = (c, d, a^2, ab, b^2).$$

Then

$$(J_{\text{dec}} : \delta) = K_{\text{dec}}. \quad (10.21)$$

Hence the local failure ring still has nilradical of exact index three, but its second intrinsic depth support is no longer reduced:

$$\mathcal{Z}_2 = V(K_{\text{dec}}). \quad (10.22)$$

Its support is the rank-two vertex, and its transverse local length is three. Thus crossing the tensor-rank-two wall changes the scheme structure of $\mathcal{Z}_2$ before it changes its set-theoretic support.

Proof. The stated normal form is obtained by taking $\ker \bar{B}_H = \mathbb{C}(h_1 \otimes w_1)$ and using

$$q_1, q_2, q_3 = \bar{B}_H(h_2 \otimes w_2)$$

as target basis. Applying $1_H \otimes T$ gives (10.20). Its maximal minors generate

$$I_3(L_{\text{dec}}) = \delta(c, d).$$

After replacing the first quadratic target column by its component modulo $\langle q_1, q_2 \rangle$, the transversality assumption allows us to normalize $\bar{C}_H(w_1^2) \equiv q_3$. The minors containing one column from each mixed group and this quadratic column give

$$\delta a^2, \quad \delta ab, \quad \delta b^2 \in J_{\text{dec}}.$$

Together with $I_3(L_{\text{dec}})$ this proves $\delta K_{\text{dec}} \subset J_{\text{dec}}$, hence $K_{\text{dec}} \subset (J_{\text{dec}} : \delta)$.

For the reverse inclusion use the transverse grading. The degree-three part of J_{dec} is exactly $\delta(c, d)$; every minor using a quadratic column has degree at least four. Modulo K_{dec} , every class is represented by

$$\lambda + \mu a + \nu b.$$

If $r\delta \in J_{\text{dec}}$, the degree-two term rules out $\lambda \neq 0$, while the degree-three identity

$$(\mu a + \nu b)\delta \in \delta(c, d)$$

forces $\mu = \nu = 0$. Thus $r \in K_{\text{dec}}$, proving (10.21).

Since $\delta \in (c, d)$, one has $\delta^2 \in \delta(c, d) \subset J_{\text{dec}}$, whereas $\delta \notin J_{\text{dec}}$ by degree. Therefore the nilradical of $P/(\delta J_{\text{dec}})$ has exact index three. Finally

$$\text{Ann}(N^2) = (\delta J_{\text{dec}} : \delta^2) = (J_{\text{dec}} : \delta) = K_{\text{dec}}.$$

The quotient by K_{dec} has basis $1, a, b$, proving the length assertion. $\square$

Proposition 10.6 (The containment locus is the finite line scheme). *For fixed $R \in G_4^\circ$, the locus*

$$\mathcal{C}_R = \{H \in \text{Gr}(2, V) : h_H \equiv 0\}$$

is canonically the Fano scheme $F_1(Y_R)$ of lines on the smooth quartic Y_R . It is finite and reduced. At its mixed-regular points the local failure ideal is (10.15), and the contact morphism $\mathcal{R}_R \rightarrow \text{Gr}(2, V)$ has fibre $\mathbb{P}^1$, rather than a finite length-four fibre.

Proof. The line of hyperplanes containing H is $\mathbb{P}(H^\perp) \subset \mathbb{P}(V^*)$. Its quartic restriction is h_H , so $h_H = 0$ identically exactly when $\mathbb{P}(H^\perp) \subset Y_R$. This identifies $\mathcal{C}_R$ with $F_1(Y_R)$.

If $L \subset Y_R$ is such a line, the normal sequence in $\mathbb{P}^3$ gives

$$\deg N_{L/Y_R} = \deg N_{L/\mathbb{P}^3} - \deg N_{Y_R/\mathbb{P}^3}|_L = 2 - 4 = -2.$$

Since N_{L/Y_R} has rank one, $N_{L/Y_R} \simeq \mathcal{O}_{\mathbb{P}^1}(-2)$, and therefore $H^0(N_{L/Y_R}) = 0$. The Zariski tangent space of the Fano scheme vanishes at every point. The Fano scheme is projective and zero-dimensional; a zero-dimensional local algebra with zero tangent space is a field by Nakayama. Hence it is finite reduced. The final two assertions follow from Theorem 10.2 and the definition of the incidence fibre. $\square$

11. PROOFS FOR THE DETERMINANTAL BOUNDARY ATLAS

We prove the assertions of Section 2. All calculations are made in the regular coefficient ring of a Schur chart. Since the identities are invariant under changes of frames, they descend by the same argument as the Fitting identities in Proposition 4.1.

11.1. A uniform fifth determinant power.

Proof of Theorem 2.1. Put $U = \text{Sym}^2 V$, so $\dim U = 10$, and regard

$$\gamma_R : U \longrightarrow S_R$$

as a fixed surjection with six-dimensional target. Choose a four-dimensional vector space K and a linear map $\rho : U \rightarrow K$ such that

$$G = (\gamma_R, \rho) : U \xrightarrow{\sim} S_R \oplus K$$

is an isomorphism. Let $M = I_H \oplus T$ and $Q = \text{Sym}^2 M$. Since $\dim V = 4$,

$$\det Q = (\det M)^5 = d^5. \quad (11.1)$$

Hence

$$\det(GQ) = \det(G)d^5.$$

Expand this 10×10 determinant by Laplace expansion along the first six rows. If $I \subset \{1, \dots, 10\}$ has cardinality six, then

$$\det(GQ) = \sum_{|I|=6} (-1)^{\epsilon(I)} \det((\gamma_R Q)_I) \det((\rho Q)_{I^c}).$$

The first determinant in every summand is a generator of $I_6(\gamma_R Q) = J$. Since $\det G$ is a unit, this proves $d^5 \in J$.

Now $\mathcal{I}_{\widehat{D}_R} = dJ$. Therefore $d^6 \in dJ$, so the nilradical generated by the class of d satisfies $N^6 = 0$. Formula (10.4) then gives $W_j = \emptyset$ for $j \geq 6$.

For the multiplication statement work locally in $A = P/(dJ)$. The class of rd^i in N^i/N^{i+1} depends on r modulo

$$(d) + (J : d^{i-1}),$$

and multiplication sends

$$(rd^i)(sd^j) = rs d^{i+j}.$$

Because

$$(J : d^{i-1}) + (J : d^{j-1}) \subset (J : d^{i+j-1}),$$

this factors through the canonical restriction to W_{i+j} . The image of $1 \otimes 1$ is the generator d^{i+j} , hence the map is surjective. The transition functions of d supply the factor $\mathcal{L}^{\otimes(i+j)}$, proving (2.3). $\square$

11.2. The first layer at projection corank two. Assume throughout this subsection that A_H is injective. After eliminating its image, the residual ideal is the ideal of 3-minors of

$$\Psi_H(T) = \left[\beta_H(1_H \otimes T), \chi_H \text{Sym}^2 T \right], \quad \dim E_H = 3. \quad (11.2)$$

Lemma 11.1 (The contact symbol without mixed regularity). *Modulo (δ) , the residual ideal of (11.2) is generated in degree four and equals*

$$h_H(u) \text{Sym}^4(\mathbb{C}_v^2)$$

under $T = uv^t$. In particular

$$J + (\delta) = (\delta, g_0, \dots, g_4)$$

with the notation of Theorem 2.3.

Proof. On the rank-one cone write $T = uv^t$. The four mixed columns are scalar multiples, with coefficients v_1, v_2 , of the two vectors

$$\beta_H(h_1 \otimes u), \quad \beta_H(h_2 \otimes u),$$

and the three quadratic columns are scalar multiples, with coefficients v_1^2, v_1v_2, v_2^2 , of the single vector $\chi_H(u^2)$. A 3-minor containing three mixed columns vanishes because the mixed span has dimension at most two. A minor containing at least two quadratic columns vanishes because those columns are proportional. Thus only minors with two mixed columns and one quadratic column survive. Their common u -factor is

$$\det[\beta_H(h_1 \otimes u), \beta_H(h_2 \otimes u), \chi_H(u^2)],$$

a binary quartic. The five possible v -monomials span $\text{Sym}^4(\mathbb{C}_v^2)$. By the same determinant comparison used in Lemma 9.4, this binary quartic is the restriction h_H of the Jacobian quartic to the pencil of hyperplanes containing H . This proves the assertion without a rank assumption on β_H . $\square$

The orbit classification in Definition 2.2 is elementary projective geometry. We record normal forms because they are also used in the colon calculation. Order the basis of $H \otimes W$ as

$$h_1w_1, h_1w_2, h_2w_1, h_2w_2.$$

Up to $GL(H) \times GL(W) \times GL(E_H)$, the mixed block can be taken to be one of the following matrices:

type	β	
3, off Q	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$	
3, on Q	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$	
2, secant	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$	
2, tangent	$\begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$	
2, H -ruling	$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$	
2, W -ruling	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$	
1, dual rank 2	$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
1, dual rank 1	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
0	0.	(11.3)

For rank two, a projective kernel line meets the smooth quadric in two reduced points, one double point, or is contained in one of the two ruling families. For rank one, the dual generator has tensor rank two or one. These alternatives are invariant and exhaustive, proving the orbit statement.

Lemma 11.2 (Exact transverse models for the nine mixed-kernel orbits). *Let*

$$P = \mathbb{C}[a, b, c, d], \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc.$$

For each normal form in (11.3) there is a nonempty Zariski-open set of quadratic blocks χ on which the Hilbert functions of

$$P/((\delta) + (J : \delta^{j-1}))$$

are those displayed in Theorem 2.3.

Proof. The two rank-three rows are Theorem 10.3 and Theorem 10.5. For the remaining rows, the assertion is a homogeneous determinant calculation. We give integer witnesses and the resulting colons; this simultaneously proves nonemptiness of the relevant rank opens.

For the secant normal form take

$$\chi = \begin{pmatrix} 1 & 2 & -2 \\ -2 & -1 & 1 \\ -2 & 1 & -2 \end{pmatrix}.$$

A direct expansion of the 3-minors followed by cancellation by δ gives

$$(J : \delta) = (c^3, c^2d, cd^2, d^3, a^2 - c^2, ab - cd, b^2 - d^2, \\ ac - 2c^2, bc - 2cd, ad - 2cd, bd - 2d^2),$$

and $(J : \delta^2) = P$. The first ideal contains δ , and the quotient has Hilbert function $(1, 4, 3)$, hence length eight.

For the tangent normal form take

$$\chi = \begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ -1 & -2 & 2 \end{pmatrix}.$$

Then

$$(J : \delta) = (c^3, c^2d, cd^2, d^3, a^2 - 10c^2, ab - 10cd, b^2 - 10d^2, \\ ac + 2c^2, bc + 2cd, ad + 2cd, bd + 2d^2),$$

again with $(J : \delta^2) = P$. The same Hilbert function $(1, 4, 3)$ follows.

For the H -ruling normal form take

$$\chi = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 2 & -1 & 0 \end{pmatrix}.$$

One obtains

$$(J : \delta) = (c^4, c^3d, c^2d^2, cd^3, d^4, 5ac^2 - c^3, 5bc^2 - c^2d, \\ 5acd - c^2d, 5bcd - cd^2, 5ad^2 - cd^2, 5bd^2 - d^3, \\ a^2 - ac, 2ab - ad - bc, b^2 - bd),$$

while

$$(J : \delta^2) = (a, b, c, d).$$

Here $\delta \notin (J : \delta)$, so $\delta^2 \notin J$; on the other hand $\delta \in (J : \delta^2)$, so $\delta^3 \in J$. After adjoining δ , the second-layer quotient has Hilbert function

$$(1, 4, 6, 4)$$

and length fifteen, while the third layer has length one.

For the W -ruling normal form take

$$\chi = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 2 \\ 1 & 0 & -2 \end{pmatrix}.$$

Then

$$(J : \delta) = (ac, bc, ad, bd, c^3, c^2d, cd^2, d^3), \quad (J : \delta^2) = P.$$

This ideal already contains δ . The Hilbert function of the second layer is

$$1, 4, 6, 4, 5, 6, 7, 8, \dots,$$

hence its Hilbert polynomial is $n + 1$ from degree four onward.

For the dual-rank-two rank-one normal form take

$$\chi = \begin{pmatrix} -2 & 1 & -1 \\ 0 & 0 & -2 \\ -2 & 0 & -1 \end{pmatrix}.$$

After adjoining δ , a homogeneous basis for the second colon may be chosen as

$$(\delta, a^2c, ac^2, a^2d, abd, b^2d, acd, ad^2, bd^2).$$

Its Hilbert function is

$$1, 4, 9, 8, 10, 12, 14, 16, \dots,$$

so the Hilbert polynomial is $2n + 2$ from degree three onward. Moreover

$$(J : \delta^2) = (a, b, c, d),$$

and the third layer is the reduced vertex.

For the dual-rank-one normal form take

$$\chi = \begin{pmatrix} -2 & 2 & -1 \\ 1 & 1 & 1 \\ 2 & 1 & -2 \end{pmatrix}.$$

A homogeneous basis after adjoining δ is

$$(\delta, a^3 + 4a^2c + 3ac^2, a^2b + 4a^2d + 3acd, \\ ab^2 + 4abd + 3ad^2, b^3 + 4b^2d + 3bd^2),$$

with Hilbert function

$$1, 4, 9, 12, 15, 18, 21, \dots$$

The next colon is, after an invertible linear change,

$$(c^2, cd, d^2, a, b),$$

so the third layer has Hilbert function $(1, 2)$ and length three.

Finally, when $\beta = 0$, surjectivity of $[\beta, \chi]$ forces χ to be invertible. Hence

$$J = I_3(\chi \operatorname{Sym}^2 T) = (\det \chi) \det(\operatorname{Sym}^2 T) = (\delta^3),$$

because $\det(\operatorname{Sym}^2 T) = \delta^3$ in dimension two. This proves the last row exactly.

All displayed Hilbert functions are computed from the displayed homogeneous bases by counting standard monomials. Their coefficient matrices have maximal rank at the given integer witnesses. The same ranks therefore hold on a nonempty Zariski-open set. Equivalently, generic flatness applied to the finitely generated graded quotients gives precisely the coefficient-transverse opens used in the statement. $\square$

Proof of Theorem 2.3. Lemma 11.1 gives the first layer and its contact interpretation. The orbit list follows from (11.3). Lemma 11.2 gives the second and third layers on each coefficient-transverse open. The nilpotency index is

$$1 + \min\{k : \delta^k \in J\}$$

by Theorem 10.1; the displayed colons give the values in the table.

If $b \leq 1$, two mixed columns never span two independent target directions, so the determinant in the proof of Lemma 11.1 vanishes identically and $h_H = 0$. If the kernel is the H -ruling $h_0 \otimes W$, one of the two mixed vectors vanishes for every u , so again $h_H = 0$. For the W -ruling, the mixed span drops only at $u = w_0$, giving the forced contact root but not identical vanishing on the transverse open. This proves the remaining assertions. $\square$

11.3. Rank-drop depth and the finite coefficient atlas.

Proof of Proposition 2.6. Eliminate the rank- a block A_H and put $q = 6 - a$. The remaining target has dimension q . A maximal residual minor uses at most b independent mixed columns, each of transverse degree one. Every remaining column must come from the quadratic block and has transverse degree two. Therefore every nonzero generator of J has degree at least

$$b + 2(q - b) = 2q - b.$$

If $2k < 2q - b$, the homogeneous polynomial δ^k cannot lie in J , proving

$$k \geq q - \lfloor b/2 \rfloor.$$

Theorem 2.1 gives $k \leq 5$.

The quadratic block has only three source dimensions when $c = 2$. Surjectivity of the original quotient after eliminating A_H therefore implies $b + 3 \geq q$. Combining this with $b \leq 4$ gives exactly the table in the statement. The equivalence between $d^k \in J$ and the augmented-rank condition $\varepsilon_k(J) = 0$ is ordinary linear algebra in the degree $2k$ homogeneous component. In the type $(a, b) = (0, 3)$ the lower bound already gives $k \geq 5$, while $d^5 \in J$, so $k = 5$ exactly. $\square$

The same coefficient construction works for $c = 3, 4$. Because $d^5 \in J$, it terminates at $k = 5$. Colon ideals are kernels of the maps

$$(P_c)_m \xrightarrow{\cdot d^{j-1}} (P_c)_{m+c(j-1)} / J_{m+c(j-1)},$$

hence are obtained by finite syzygy elimination. Noetherianity gives finite homogeneous generating sets. Rank stratification of the resulting coefficient matrices, followed by the usual flattening stratification of the finitely presented graded quotients, gives the finite atlas in Definition 2.4. This is the finite algebraic replacement for an unspecified infinite colon tower.

11.4. Projection corank three.

Lemma 11.3 (No cubic determinant power at corank three). *At projection corank three, $d^3 \notin J$.*

Proof. Here $\dim W = 3$ and $\deg d = 3$. First suppose A_H has rank one. After eliminating its image the residual target has dimension five. The maximal-rank mixed case has $\text{rank } \overline{B}_H = 3$; every rank drop gives a subrepresentation of this case and cannot create a determinant representation that was absent before.

The degree-seven, degree-eight and degree-nine symbols of the five-minors use respectively three, two and one mixed columns. As right $GL(W)$ -representations they are contained in

$$\bigwedge^3 W \otimes \bigwedge^2 \text{Sym}^2 W, \quad \bigwedge^2 W \otimes \bigwedge^3 \text{Sym}^2 W, \quad W \otimes \bigwedge^4 \text{Sym}^2 W.$$

The elementary Schur decompositions are

$$\bigwedge^2 \text{Sym}^2 W = \mathbb{S}_{(3,1,0)} W,$$

$$\bigwedge^3 \text{Sym}^2 W = \mathbb{S}_{(4,1,1)} W \oplus \mathbb{S}_{(3,3,0)} W,$$

and

$$\bigwedge^4 \text{Sym}^2 W = \mathbb{S}_{(4,3,1)} W.$$

Consequently the degree-eight symbol is a sum of

$$\mathbb{S}_{(5,2,1)}, \mathbb{S}_{(4,4,0)}, \mathbb{S}_{(4,3,1)}, \mathbb{S}_{(4,2,2)},$$

and the degree-nine symbol is a sum of

$$\mathbb{S}_{(5,3,1)}, \mathbb{S}_{(4,4,1)}, \mathbb{S}_{(4,3,2)}.$$

The degree-seven symbol is $\mathbb{S}_{(4,2,1)}$. To obtain a degree-nine element of the ideal one may multiply the degree-seven symbol by degree-two coordinate functions, the degree-eight symbol by degree-one coordinate functions, or take the degree-nine symbol itself. None can contain

$$(\det W)^3 = \mathbb{S}_{(3,3,3)} W :$$

every displayed highest weight has first part at least four, and the Littlewood–Richardson rule only adds boxes. Since d^3 spans this determinant representation, it is not in J .

If $A_H = 0$, the residual target has dimension six. A degree-nine minor can occur only when the mixed block has rank three and uses all three mixed columns and three quadratic columns. Its right type is contained in

$$\det W \otimes \bigwedge^3 \text{Sym}^2 W = \mathbb{S}_{(5,2,2)} W \oplus \mathbb{S}_{(4,4,1)} W,$$

again with no $(3,3,3)$ summand. If the mixed rank drops, the minimal degree is larger than nine. This proves the assertion in all cases. $\square$

Proof of Theorem 2.7. Lemma 11.3 gives $d^3 \notin J$, while Theorem 2.1 gives $d^5 \in J$. Therefore the first determinant power in J is d^4 or d^5 . Definition 2.4 says precisely that $d^4 \in J$ when $\varepsilon_4(J) = 0$. The nilpotency index is one plus this first exponent. $\square$

11.5. Projection corank four. We use one standard plethysm calculation (see, for example, [18] for Schur functors and character methods). The identity may also be checked directly by the Weyl character formula:

$$\bigwedge^6 \text{Sym}^2 V = \mathbb{S}_{(5,4,2,1)} V \oplus \mathbb{S}_{(4,4,4,0)} V. \quad (11.4)$$

Both summands occur with multiplicity one.

Lemma 11.4 (The Jacobian projection). *Under*

$$\bigwedge^6 \text{Sym}^2 V \simeq (\det V)^5 \otimes \left(\bigwedge^4 \text{Sym}^2 V \right)^*,$$

the summand $\mathbb{S}_{(4,4,4,0)} V$ is dual, up to the displayed determinant twist, to

$$\det V \otimes \text{Sym}^4 V \subset \bigwedge^4 \text{Sym}^2 V.$$

The projection

$$\bigwedge^4 \text{Sym}^2 V \longrightarrow \det V \otimes \text{Sym}^4 V$$

sends $q_1 \wedge \cdots \wedge q_4$, up to a nonzero scalar, to

$$\det \left(\frac{\partial q_i}{\partial x_j} \right).$$

Proof. Duality of Schur functors gives

$$(\det V)^5 \otimes \mathbb{S}_{(4,4,4,0)} V^* \simeq \det V \otimes \text{Sym}^4 V.$$

The Jacobian determinant is alternating in the four quadrics and is $GL(V)$ -equivariant with exactly this target weight. It is nonzero, for instance on $x_1^2, x_2^2, x_3^2, x_4^2$. Multiplicity one in (11.4) therefore identifies it with the stated projection. $\square$

Proof of Theorem 2.8. At $H = 0$ the ideal J is generated in degree twelve by the 6-minors of $\gamma_R \text{Sym}^2 T$. Under right multiplication of T , their span is a quotient of the representation $\bigwedge^6 \text{Sym}^2 V$. The polynomial d^3 spans $(\det V)^3 = \mathbb{S}_{(3,3,3,3)} V$, which does not occur in (11.4). Hence $d^3 \notin J$.

Let

$$\lambda_R = \bigwedge^6 \gamma_R.$$

By Lemma 11.4, the component of λ_R on the $\mathbb{S}_{(4,4,4,0)}$ -summand is nonzero exactly when the Jacobian quartic F_R is nonzero. For $R \in G_4^\circ$, the quartic $Y_R = V(F_R)$ is smooth, so this component is nonzero.

The degree-four coordinate ring of $\text{End}(V)$ contains the right representation $\mathbb{S}_{(4,0,0,0)} V = \text{Sym}^4 V$. Moreover

$$\mathbb{S}_{(4,4,4,0)} V \simeq (\det V)^4 \otimes (\text{Sym}^4 V)^*.$$

Hence the unique equivariant map

$$\mathbb{S}_{(4,4,4,0)} V \otimes \text{Sym}^4 V \longrightarrow (\det V)^4$$

is, up to scalar, the evaluation pairing. The copy of $\mathbb{S}_{(4,4,4,0)} V$ in J_{12} is nonzero precisely by the preceding Jacobian calculation. Choose a degree-four matrix coefficient whose $\text{Sym}^4 V$ -component pairs nontrivially with that copy. Multiplication therefore has nonzero projection onto $(\det V)^4$.

By the Cauchy decomposition of the degree-sixteen coordinate ring of $\text{End}(V)$, the right $(\det V)^4$ -isotypic component has multiplicity $\dim(\det V^*)^4 = 1$; it is the line spanned by d^4 . Consequently $d^4 \in J$. Together with $d^3 \notin J$, this gives nilpotency index five. $\square$

11.6. Specialization and associated supports. The atlas is compatible with specialization in the following precise sense. Over any finite-type coefficient base carrying A_H, B_H, C_H , the generators of J , the matrices M_k , and the syzygy matrices computing $J : d^{j-1}$ have polynomial entries in the base coordinates. Rank conditions therefore define locally closed determinantal strata. On a stratum where the finitely presented graded modules

$$P_c / ((d) + (J : d^{j-1})), \quad 1 \leq j \leq 5,$$

and the cokernels of (2.3) are flat, their Hilbert polynomials and multiplication ranks are constant and commute with base change. Further stratifying by the Fitting ideals of these modules and by Noetherian induction gives a finite locally closed refinement on which the associated supports have fixed incidence type. Thus a collision or rank drop is recorded by an actual determinantal specialization of the graded algebra, rather than by a change of labels in a generic primary decomposition.

12. DEFORMATION TO THE NORMAL CONE: STANDARD MECHANISM AND THE COMPUTED SPECIAL FIBRE

The passage from an ideal to its extended Rees algebra is the standard deformation-to-the-normal-cone construction; see [17, §5.1] and [19, Tag 062Z]. We use it here only as the canonical mechanism that packages the manuscript-specific graded algebra computed in Sections 10 and 11.

Proposition 12.1 (Intrinsic deformation to the normal cone). *Let $\widehat{\mathcal{N}}_R$ be the intrinsic nilradical of $\widehat{D}_R$. On an affine open $\text{Spec } A \subset \widehat{D}_R$, with nilradical $N \subset A$, put*

$$\mathcal{R}_N = A[t, Nt^{-1}] \subset A[t, t^{-1}].$$

These algebras glue and define a flat family

$$\mathfrak{D}_R \longrightarrow \mathbb{A}^1.$$

Over $\mathbb{G}_m$ it is $\hat{D}_R \times \mathbb{G}_m$, while its special fibre is

$$\mathfrak{D}_{R,0} \simeq \operatorname{Spec} \hat{\Delta}_R \operatorname{gr}_{\hat{N}_R} \mathcal{O}_{\hat{D}_R} = C_{\hat{\Delta}_R/\hat{D}_R}.$$

The construction is functorial for isomorphisms of the abstract failure scheme.

Proof. The extended Rees algebra is the standard construction. Multiplication by t is injective because $\mathcal{R}_N$ is a subring of $A[t, t^{-1}]$. Hence it is torsion-free over the PID $\mathbb{C}[t]$, and therefore flat. Inverting t gives $A[t, t^{-1}]$. Modulo t , the class of nt^{-j} , $n \in N^j$, depends only on $n \bmod N^{j+1}$, giving

$$\mathcal{R}_N/(t) \simeq \bigoplus_{j \geq 0} N^j/N^{j+1}.$$

Localization compatibility glues the construction. Since N is the ideal of the reduction $\hat{\Delta}_R$ in $\hat{D}_R$, the special fibre is the normal cone. The nilradical is intrinsic, so the construction is functorial. $\square$

The point of Revision 127 is not Proposition 12.1 itself. The new content is that its special fibre is now bounded and stratified:

$$\operatorname{gr}_{\hat{N}_R} \mathcal{O}_{\hat{D}_R} = \mathcal{O}_{\hat{\Delta}_R} \oplus \bigoplus_{j=1}^5 \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

with the quotient multiplication (2.3). Projection corank two has the mixed-kernel atlas of Theorem 2.3; all A_H -rank drops have the finite valuation bounds of Proposition 2.6; projection corank three has the exact index-five/index-six dichotomy of Theorem 2.7; and projection corank four has exact index five by Theorem 2.8.

Flattening is used only to organize coefficient specializations of these explicitly presented finite modules. We use the standard flattening-stratification formalism in [20, Tag 052F]; no novelty is claimed for that formalism either.

Remark 12.2 (Novelty boundary). The standard ingredients in this section are the extended Rees algebra, deformation to the normal cone, and flattening stratification. The manuscript-specific statements are the Fitting factorization, the global determinant-power bound $d^5 \in J$, the nine mixed-kernel normal forms and their graded supports, the corank-three Schur-type exclusion, the corank-four Jacobian-covariant calculation, the collision primary laws, and the polarized reconstruction theorem. These roles are kept separate in the statements and in the referee response.

13. THE CLASSICAL INCIDENCE SURFACE AND THE NEW INVARIANT

This section distinguishes three geometric objects associated with the same relation system. Let $e = 4$ and let $q_i(x, y)$ denote the symmetric bilinear polarization of $q_i(x)$, normalized by $q_i(x, x) = q_i(x)$. Define

$$Z_R = \{([x], [y]) \in \mathbb{P}(V^*) \times \mathbb{P}(V^*) : q_i(x, y) = 0 \text{ for all } i\}. \quad (13.1)$$

Proposition 13.1. *For $R \in G_4^\circ$, projection on the first factor identifies Z_R with Y_R . Transposition of the two factors induces a fixed-point-free involution on this K3 surface. The polarization recovered in Theorem 1.1 is the first projection's quartic polarization.*

Proof. The bilinear equations at x are $J_R(x)y = 0$, up to the common nonzero factor two. Off Y_R there is no projective solution; on Y_R the matrix has rank exactly three because Y_R is smooth. Inverting a nonzero three-minor gives a unique kernel line, with its scheme structure, and these kernel lines glue to an inverse for the projection $Z_R \rightarrow Y_R$. The equations are symmetric in x, y , so transposition acts. A fixed point would have $[x] = [y]$ and hence $q_i(x) = 0$ for every i , contrary

to basepoint-freeness. The final assertion follows from the first projection and its hyperplane bundle. $\square$

The discriminant of the web is a different embedded hypersurface. If $q_i(x) = x^t A_i x$, its equation is

$$\det\left(\sum_{i=0}^3 \lambda_i A_i\right) = 0 \quad \text{in } \mathbb{P}(R), \quad (13.2)$$

whereas the ramification equation is $\det J_R(x) = 0$ in $\mathbb{P}(V^*)$. The former varies the quadric in the web; the latter varies a point of the source of the quadratic morphism. The two determinants must not be identified simply because both have degree four. The classical construction recalled in [13, §4] relates the bilinear K3 surface to the resolution of the web's symmetroid and to a nodal Enriques quotient; see also the account of Reye congruences in [9]. Proposition 13.1 places our ramification surface in that construction by an explicit kernel-line isomorphism. No claim is made that the existence of that K3 surface, its involution, or webs of quadrics is new.

The invariant used here starts instead with the abstract multiplication-failure scheme. It obtains E as the annihilator of its nilradical, constructs $\omega_E \otimes \mathcal{N}^{-9}$ without the web or either projective space, and recovers the section ring of $\mathcal{O}_{Y_R}(4)$. The corank-two theorem further recovers an additional associated support from the square of the nilradical. These are statements about nonreduced multiplication-failure schemes, not merely about the classical incidence surface or the discriminant cycle.

14. REYE GEOMETRY, INFINITESIMAL MODULI, AND THE NOVELTY BOUNDARY

We now separate three statements that were too close together in the preceding revision: the classical moduli of webs and Reye congruences, the exact differential calculation for the polarized K3 map, and the nonreduced reconstruction theorem.

14.1. The classical nine-dimensional quotient. For $e = 4$, a relation space $R \subset \text{Sym}^2 V$ is a web of quadrics. The bilinear incidence surface Z_R of §13 is a K3 surface with its fixed-point-free factor-exchange involution; its quotient is the classical Reye Enriques surface associated with the web. This construction and the resulting congruence of lines are classical; see [7, 1, 10, 13, 9, 16].

Arrondo–Sols compute that the Hilbert scheme of Reye congruences is smooth of dimension 24 and rational, and describe it by an open Grassmannian parameter of the corresponding data. After quotienting by the 15-dimensional projective coordinate group, the expected moduli dimension is therefore nine. In characteristic zero the nodal Enriques locus is the classical nine-dimensional divisor in the ten-dimensional Enriques moduli space; the modern theorem [16] identifies every classical nodal Enriques surface as a Reye congruence.

Thus the equality

$$24 - 15 = 9$$

is not a novelty claim of this paper.

Proposition 14.1 (The classical tangent quotient). *At a general regular web with finite projective stabilizer, the tangent space to the Reye moduli locus is the quotient of the 24-dimensional web tangent space by the 15-dimensional infinitesimal $\text{PGL}(V)$ -orbit:*

$$T_{[R]}\mathcal{M}_{\text{Reye}} \simeq T_R \text{Gr}(4, \text{Sym}^2 V) / \mathfrak{pgl}(V), \quad \dim T_{[R]}\mathcal{M}_{\text{Reye}} = 9.$$

The differential of the polarized-K3 map of Theorem 1.4 factors through this quotient.

Proof. The first statement is the tangent-space form of the smooth 24-dimensional Hilbert calculation of Arrondo–Sols on the regular web locus, followed by the infinitesimal quotient by projective coordinates. The stabilizer is finite at a general point, so the orbit tangent has dimension 15.

The construction $R \mapsto Z_R \simeq Y_R$ is equivariant for $\mathrm{PGL}(V)$. Passing from an embedded quartic to its polarized isomorphism class kills the infinitesimal projective orbit; hence the differential factors through the displayed quotient. $\square$

Proposition 14.2 (Web parameter and Reye Hilbert point). *Let*

$$\mathcal{G} = \mathrm{Gr}(4, \mathrm{Sym}^2 V)$$

and let $\mathcal{G}^{\mathrm{reg}}$ be the open on which the classical Reye construction has the expected Hilbert polynomial. The assignment of a web to its Reye congruence is a $\mathrm{PGL}(V)$ -equivariant morphism

$$\Phi : \mathcal{G}^{\mathrm{reg}} \longrightarrow \mathrm{Hilb}_{\mathrm{Reye}}.$$

Under the Grassmannian parameterization of the regular Reye locus in Arrondo–Sols, Φ identifies a dense open of $\mathcal{G}^{\mathrm{reg}}$ with the corresponding 24-dimensional Hilbert component. In particular, at a general regular web,

$$d\Phi_R : \mathrm{Hom}(R, \mathrm{Sym}^2 V/R) \xrightarrow{\sim} T_{[\mathrm{Reye}(R)]} \mathrm{Hilb}_{\mathrm{Reye}}.$$

After quotienting by the infinitesimal projective orbit this gives the tangent quotient in Proposition 14.1.

Proof. Over $\mathcal{G}$ the universal rank-four subbundle of $\mathrm{Sym}^2 V$ gives the universal web. The incidence equations defining the Reye congruence are polynomial in the universal Pluecker coordinates. Restrict to the open on which their Hilbert polynomial is the regular Reye polynomial. Flatness there and the Hilbert functor give the morphism Φ ; equivariance is immediate from the construction.

Arrondo–Sols describe the regular Reye Hilbert component by the open Grassmannian parameter of the defining four-dimensional system of quadrics. With the present convention this parameter is precisely a dense open of $\mathrm{Gr}(4, \mathrm{Sym}^2 V)$ (duality only changes whether one writes the system or its annihilator). Their smoothness and dimension-24 statement therefore identifies the tangent space of the Hilbert point with the tangent space of this Grassmannian. The latter is $\mathrm{Hom}(R, \mathrm{Sym}^2 V/R)$, proving the displayed isomorphism. Since the construction is $\mathrm{PGL}(V)$ -equivariant, the orbit tangent maps to the orbit tangent and the quotient gives Proposition 14.1. $\square$

The role of Lemma 7.1 is now precise. It is not used to discover the number nine. Its exact 25×25 integral minor proves that the particular polarized K3 attached to our multiplication problem does not collapse any of the 24 embedded web directions at the explicit rational point. After projective quotient, it proves rank nine for the map to polarized K3 moduli. Since maximal rank is open, the induced map from the classical nine-dimensional Reye deformation space to our polarized K3 image is generically unramified. This is the conceptual tangent statement behind Theorem 1.4; the determinant is an exact arithmetic witness for it.

14.2. What the finite ambiguity means. The preceding paragraph also locates the remaining Torelli ambiguity. The failure scheme first reconstructs the polarized K3 $(Y_R, \mathcal{O}_{Y_R}(1))$. A relation system additionally supplies the free involution coming from factor exchange on the bilinear incidence model and hence an Enriques quotient with its Reye data. Classical Reye geometry then has substantial rigidity. In particular, Dolgachev–Reider prove that for a Reye polarization on an Enriques surface the stable rank-two bundle with c_1 equal to that polarization and $c_2 = 3$ is the Reye bundle and is unique up to isomorphism; it is the restriction of the universal quotient bundle from $G(2, 4)$.

Consequently the generic finite fibre in Theorem 1.4 should not be read as an uncontrolled family of quadratic tensors. Once a compatible Enriques quotient and Reye polarization are fixed, the Grassmannian Reye bundle is rigid in the preceding sense. The remaining finite packet is concentrated in the finite choices of compatible Reye/Enriques data over the recovered polarized

K3, together with the passage back to the web. The present paper proves finiteness but does not claim a numerical degree or generic degree one; such a claim would require a separate enumeration of those compatible involutions and web structures.

There is a second useful localization. On the mixed-regular projection-corank-two locus, the unsplit contact quartic of §9 is the restriction of the already recovered quartic Y_R to the pencil of hyperplanes containing H . Thus the generic corank-two contact cover itself is determined by the polarized K3. The following theorem turns the finite Torelli packet into a precise finite-cover statement and identifies its monodromy.

Theorem 14.3 (Generic Torelli packet as a finite etale monodromy cover). *Let $\mathcal{W}$ be the irreducible nine-dimensional coarse quotient of the regular web locus by projective equivalence on the finite-stabilizer open, and let $\mathcal{K}$ be the nine-dimensional polarized-K3 image of Theorem 1.4. There are nonempty opens*

$$\mathcal{W}^\circ \subset \mathcal{W}, \quad \mathcal{K}^\circ \subset \mathcal{K}$$

such that

$$\tau : \mathcal{W}^\circ \longrightarrow \mathcal{K}^\circ$$

is finite etale of a constant degree

$$d_{\text{Tor}} = [\mathbb{C}(\mathcal{W}) : \mathbb{C}(\mathcal{K})].$$

Its geometric monodromy is transitive on the d_{Tor} points of a fibre. Equivalently, if L is the normal closure of $\mathbb{C}(\mathcal{W})/\mathbb{C}(\mathcal{K})$, the monodromy group is the permutation image of

$$\text{Gal}(L/\mathbb{C}(\mathcal{K}))$$

on the embeddings of $\mathbb{C}(\mathcal{W})$ in L .

The degree-four contact incidence and its discriminant are pulled back from the polarized quartic on $\mathcal{K}^\circ$. Hence they are constant on a Torelli fibre and do not account for d_{Tor} ; any further separation of the finite packet must use web-dependent structure beyond the polarized quartic, such as the exceptional mixed-kernel or deeper projection-corank strata.

Proof. The source $\mathcal{W}$ is irreducible because it is obtained from a nonempty open of the irreducible Grassmannian $\text{Gr}(4, \text{Sym}^2 V)$ by the projective-coordinate quotient on the finite-stabilizer locus. Theorem 1.4 says that the dominant map $\tau : \mathcal{W} \dashrightarrow \mathcal{K}$ is generically finite. Thus the induced extension of function fields is finite. In characteristic zero it is separable.

Normalize a nonempty affine open of $\mathcal{K}$ in $\mathbb{C}(\mathcal{W})$. After deleting the complement of the locus on which the original rational map is defined, and then deleting the branch and nonflat loci, one obtains a finite etale morphism from a nonempty open of $\mathcal{W}$. Its degree is the function-field degree displayed above. Connectedness of the generic source, or equivalently irreducibility of $\mathcal{W}$, gives transitivity of geometric monodromy. The description by the normal closure is the standard field-theoretic realization of that action.

The contact incidence is defined solely from the recovered embedded quartic $Y \subset \mathbb{P}^3$ by

$$(H, [\alpha]) \quad \text{with} \quad H \subset \ker \alpha,$$

and the discriminant is the discriminant of its restricted binary quartic. Both constructions commute with isomorphism of the polarized quartic. They therefore descend to $\mathcal{K}^\circ$ and pull back along τ , proving the last assertion. $\square$

14.3. Comparison with nonreduced reconstruction theory. The use of nilpotent layers has classical antecedents. A ribbon is a square-zero double structure with an invertible conormal module, and Bayer–Eisenbud developed its intrinsic and canonical geometry [4]. Manolache surveys Cohen–Macaulay nilpotent structures and their filtrations [15]. Primitive multiple schemes carry a canonical filtration by lower multiplicities in the sense of Drézet [11].

Our local corank-one model

$$t(t, f) = (t) \cap (t, f)^2$$

indeed has a square-zero nilradical line on the embedded support, and the corank-two model has one further nonzero nilpotent power. We do not claim that the existence of a nilpotent filtration is new. The specific contribution is the following chain of implications, which does not occur in the cited multiple-structure theory:

$$\text{multiplication Fitting scheme} \implies (\mathcal{Z}_1, \widehat{\mathcal{N}}|_{\mathcal{Z}_1}) \implies \omega_{\mathcal{Z}_1} \otimes \widehat{\mathcal{N}}^{-9} \implies (Y_R, \mathcal{O}_{Y_R}(1)),$$

together with

$$\mathcal{Z}_2 = \text{rank-two Schubert support}$$

on the mixed-regular higher-corank locus. The ambient failure scheme is not assumed to be a primitive multiple scheme, and the filtration is defined by annihilators of powers of its intrinsic nilradical rather than by choosing an ambient thickening.

14.4. Ballico's failure-locus framework. The complete theorem text of Ballico's 1993 article [2] was not available to this revision. We therefore do not infer either anticipation or non-anticipation from its title or metadata. A theorem-level comparison is possible with Ballico's accessible 1996 continuation [3]. There the basic data are an integral projective variety X , a very ample line bundle L , the complete embedding by $H^0(X, L)$, and multiplication maps such as

$$H^0(L) \otimes H^0(L^k) \longrightarrow H^0(L^{k+1}).$$

The principal results propagate failure of quadratic or higher normality to suitable linear sections and finite subschemes; the parameter Grassmannian is a Grassmannian of projective linear sections.

The present construction has a different input and a different scheme-theoretic output. The fixed object is the finite local cube-zero algebra B_R ; a point of X_R is a unital subspace $\mathbb{C}1 \oplus K \subset B_R$; and the failure scheme is the zeroth Fitting scheme of

$$\text{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \longrightarrow B_R \otimes \mathcal{O}.$$

Its embedded associated supports, nilradical powers, and the polarized K3 reconstructed from their line bundles are part of the statement. Ballico's 1996 theorems are therefore an important precedent for organizing multiplication failure over a Grassmannian, but they do not by themselves supply the Fitting normal forms or the nilpotent-depth reconstruction proved here. This comparison is deliberately limited to the theorem text actually inspected; the separate 1993 comparison remains documentary work rather than a premise in a novelty claim.

14.5. Residue of the main theorem. After these classical comparisons, the claims that remain specific to this paper are concise.

First, an abstract multiplication-failure scheme reconstructs the quartic polarization by the intrinsic line $\omega_{E_R} \otimes \mathcal{N}_R^{-9}$ and its section ring. Second, the same nonreduced scheme has a functorial annihilator filtration whose next nonzero stage recovers the first higher-corank Schubert support. Third, on the mixed-regular rank-two locus the unsplit contact ideal has a complete collision specialization law, the fifth-power vertex component remains genuinely primary, and the associated graded nilpotent algebra identifies both the contact cone and the rank-two Schubert support. The first decomposable mixed-kernel wall is also computed scheme-theoretically. Finally, Theorem 14.3 identifies the generic inverse problem with a finite étale monodromy cover. These statements are independent of whether the nine-dimensional Reye family itself is classical.

APPENDIX A. WEIGHTED DETERMINANTAL FLAGS AND THE CLASSICAL ORDER PROFILE

We record explicitly how the retained weighted-matrix calculation fits the classical determinantal theory. This comparison concerns the generic matrix only; the fixed-tensor theorem of §8 was proved independently.

Let $A = (a_{ij})$ be a generic $q \times p$ matrix over a characteristic-zero field k , with $p \geq q$, and put $R_0 = k[a_{ij}]$, $I_0(A) = R_0$. For a nonincreasing sequence $\rho = (r_1, \dots, r_s)$ write

$$I^\rho = \prod_{\nu=1}^s I_{r_\nu}(A), \quad \gamma_j(\rho) = \sum_{\nu=1}^s \max(r_\nu - j + 1, 0).$$

The classical formula is

$$I^\rho = \bigcap_{j=1}^{r_1} I_j(A)^{(\gamma_j(\rho))}. \quad (\text{A.1})$$

Here the superscript in parentheses denotes symbolic power. See De Concini–Eisenbud–Procesi [8], and its statement as Theorem 2.4 in Bruns–Conca [5], together with Proposition 2.2 there for the symbolic-order basis. The characteristic hypotheses and attribution are stated on printed pages 13–14 of that exposition.

Proposition A.1 (Coefficientwise suspension). *In $R_0[t]$, put*

$$F = t \sum_{i=0}^q t^{q-i} I_i(A), \quad P_j = (t, I_j(A)).$$

Then

$$F = (t) \cap \bigcap_{j=1}^q P_j^{(q-j+2)}. \quad (\text{A.2})$$

This is an irredundant primary decomposition. At a point with $t = 0$ where A has rank a , the nilradical of the corresponding local quotient has index $q - a + 1$.

Proof. In the regular local ring $(R_0)_{I_j}$, the symbolic order of an i -minor is $\max(i - j + 1, 0)$. Adjoining t gives the coefficient rule

$$\sum_{u \geq 0} t^u f_u \in P_j^{(b)} \iff f_u \in I_j^{(\max(b-u, 0))} \text{ for every } u. \quad (\text{A.3})$$

For precision, the associated graded ring of the maximal ideal $(t, I_j(R_0)_{I_j})$ is a polynomial ring over its residue field; its homogeneous t -degrees are independent. The order of the displayed sum is therefore the minimum of $u + \text{ord}_{I_j}(f_u)$. Contracting the maximal-ideal power proves (A.3), including the assertion that each contracted symbolic power is primary.

Intersecting with (t) deletes coefficient degree zero. For $1 \leq u \leq q$, put $k = q + 1 - u$. The remaining coefficient conditions on the right of (A.2) are

$$f_u \in \bigcap_{j=1}^k I_j^{(k-j+1)} = I_k,$$

the one-factor case $\rho = (k)$ of (A.1). For $u \geq q + 1$ there is no condition. These are exactly the coefficients of F , proving the equality.

For irredundancy, let d_{j-1} be a fixed $(j-1)$ -minor, with $d_0 = 1$. The element $t^{q-j+1} d_{j-1}^2$ belongs to every component other than $P_j^{(q-j+2)}$. Indeed, its order along P_j is $q - j + 1$, while at P_k for $k < j$ its order is $q - j + 1 + 2(j - k) \geq q - k + 2$, and for $k > j$ its t -order suffices. The element d_q^2 witnesses the necessity of (t) . The associated primes are therefore exactly the displayed radicals.

After inverting an a -minor and Schur reducing, the localized coefficient ideal contains t^{q-a+1} , whereas t^{q-a} has coefficient one in degree $q-a$, where membership would require a one-minor of the residual matrix. That coefficient ideal is proper at the point. Thus the radical (t) has the asserted exact index. $\square$

The exponent $q-j+2$ is therefore the scalar-variable suspension of the classical minor order $k-j+1$, taken coefficientwise. This is a direct consequence of the classical profile plus (A.3); it is not a new straightening theorem. The presentation $I_q[A, tI_q] = \sum_i t^{q-i} I_i(A)$ can also be viewed as substituting a scalar identity block in a larger generic matrix. Such substitution alone does not justify preservation of primary components or flatness. The coefficient proof is what supplies the exact intersection here, while the ideal $\delta\mathbf{m} + (g_0, \dots, g_4)$ in §8 requires its separate fixed-tensor analysis.

APPENDIX B. AN EXPLICIT SYSTEM AND EXACT CERTIFICATE SPECIFICATIONS

The rational system used throughout is the following. It is the same relation system as in the preceding manuscript's exact smoothness example, now used for a differential and a higher-corank calculation.

$$\begin{aligned} q_0 &= x_0^2 + 2x_0x_1 - 2x_0x_2 + 2x_0x_3 + x_1x_2 + x_1x_3 + x_2x_3 \\ q_1 &= -x_0x_1 - x_0x_3 + x_1^2 - x_1x_2 - x_1x_3 - 2x_2x_3 \\ q_2 &= 2x_0x_1 - x_0x_2 - 2x_0x_3 + 2x_1x_2 + x_2^2 + x_2x_3 \\ q_3 &= x_0x_2 + x_1x_2 - 2x_2x_3 + x_3^2 \end{aligned}$$

For a compact complete specification of its quartic, list the 35 degree-four exponent tuples in descending lexicographic order, starting with $(4, 0, 0, 0)$ and ending with $(0, 0, 0, 4)$. The coefficients of $F = \det(\partial q_i / \partial x_j)$ in that order are the rows of

$$\begin{pmatrix} -8 & -4 & -20 & 16 & 20 & 0 & -28 \\ -2 & -36 & 40 & 20 & -50 & -20 & -26 \\ 72 & -6 & 0 & -66 & -6 & 30 & 4 \\ -14 & -4 & -6 & -12 & 28 & 18 & -56 \\ 28 & 6 & -18 & 8 & -2 & 16 & 2 \end{pmatrix}.$$

Thus this display, together with the fixed monomial order, specifies every coefficient, including zeros. Alternatively, the determinant of the displayed four quadrics is its coordinate-free recipe.

For any list of homogeneous forms $f_1, \dots, f_s$ define the degree- d Macaulay coefficient matrix $\mathcal{M}_d(f)$ as follows. Rows are all degree- d monomials in the same descending lexicographic order. Columns are the forms $x^\beta f_i$, with i outermost and all β of degree $d - \deg f_i$ in descending lexicographic order innermost. The entry in row α is the coefficient of x^α . All entries for this example are integers.

Matrix	Size	Certified rank	Minor modulo 101
$[F, G_{01}, \dots, G_{36}]$	35×25	25	41
$\mathcal{M}_5(q_0, q_1, q_2, q_3)$	56×80	56	44
$\mathcal{M}_9(F_{x_0}, F_{x_1}, F_{x_2}, F_{x_3})$	220×336	220	79

The differential minor consists of the first 25 rows and all columns; its full integer value is (7.4). The following deterministic rule specifies the other two minors without printing large matrices. Reduce the displayed coefficient matrix modulo 101. Scan columns from left to right. At the next pivot step choose the first remaining row with nonzero entry, exchange it with the next pivot row, normalize that row, and eliminate below. Skip a column if there is no nonzero remaining entry.

Record the original row indices in the order selected and the pivot columns. The square submatrix of the *original*, unreduced matrix with exactly these row and column orders has the residue in the final column of the table. Full recorded index lists, along with the coefficient-generating program, are included in `evidence/K3_CERTIFICATES.json` and `checks/exact_k3.py`.

Every step is finite-field arithmetic on integer data. A nonzero residue of a specified square minor proves that integer minor is nonzero, so it proves a rank statement over $\mathbb{Q}$, not merely a rank in one numerical specialization. The Macaulay rank statements imply the homogeneous ideal containments used in Lemma 7.1. The exact integer determinant is also recomputed independently with rational-domain elimination. No floating-point rank, tolerance, or unverified smoothness inference is used.

For the corank-two quotient in (8.1), one may reproduce its quotient map by multiplying the six-dimensional quadratic quotient by

$$\begin{pmatrix} 0 & -1/2 & -1 & 1 & 0 & 0 \\ 0 & -1/2 & -1 & 0 & 1 & 0 \\ 0 & -3/2 & -2 & 0 & 0 & 1 \end{pmatrix}.$$

This matrix annihilates the three columns of $\mathrm{Sym}^2 H$ and has rank three. It yields the displayed B_H, C_H and the quartic (8.2) by ordinary multiplication. The factor 2 in the coefficient of uv is essential: it records the square of $ux_2 + vx_3$, not a divided-power convention.

An additional exact regression test uses the normalized matrix (8.3) with

$$C = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1/2 & -3 \end{pmatrix}.$$

Then $h(u, v) = uv(u-v)(u-2v)$, so the four ideals are (a, b) , (c, d) , $(a-c, b-d)$, and $(a-2c, b-2d)$. Elimination computes both sides of (1.10) over $\mathbb{Q}$, and mutual Gröbner reduction verifies their equality. It also checks that δ^2 is nonzero, that every element of $\mathfrak{m}\delta^2$ vanishes, and that $\delta^3 = 0$. This test checks a concrete instance; the general coefficient-ring proof is Lemma 8.4 and Theorem 8.5, not an extrapolation from that instance.

REFERENCES

- [1] E. Arrondo and I. Sols, *On congruences of lines in the projective space* (Chapter 6 written in collaboration with M. Pedreira), *Mém. Soc. Math. France (N.S.)* **50** (1992), 1–96. doi:10.24033/msmf.363.
- [2] E. Ballico, On the failure locus of higher order properties of embeddings in projective spaces, *Math. Nachr.* **163** (1993), 5–13. doi:10.1002/mana.19931630102.
- [3] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–313.
- [4] D. Bayer and D. Eisenbud, Ribbons and their canonical embeddings, *Trans. Amer. Math. Soc.* **347** (1995), 719–756. doi:10.1090/S0002-9947-1995-1273472-3.
- [5] W. Bruns and A. Conca, Gröbner bases and determinantal ideals, in *Commutative Algebra, Singularities and Computer Algebra*, NATO Sci. Ser. II Math. Phys. Chem. 115, Kluwer, 2003, 9–66; author preprint arXiv:math/0302058.
- [6] W. Bruns and U. Vetter, *Determinantal Rings*, Lecture Notes in Mathematics 1327, Springer, 1988.
- [7] F. Cossec, Reye congruences, *Trans. Amer. Math. Soc.* **280** (1983), 737–751. doi:10.1090/S0002-9947-1983-0716848-4.
- [8] C. De Concini, D. Eisenbud, and C. Procesi, Young diagrams and determinantal varieties, *Invent. Math.* **56** (1980), 129–165. doi:10.1007/BF01392548.
- [9] I. Dolgachev and S. Kondō, *Enriques Surfaces II*, Springer, 2025, chapter *Reye Congruences*, 67–141. doi:10.1007/978-981-96-1513-1_2.
- [10] I. Dolgachev and I. Reider, On rank 2 vector bundles with $c_1^2 = 10$ and $c_2 = 3$ on Enriques surfaces, in *Algebraic Geometry*, Lecture Notes in Mathematics 1479, Springer, 1991, 39–49. doi:10.1007/BFb0086262.
- [11] J.-M. Drézet, Primitive multiple schemes, *Eur. J. Math.* **7** (2021), 985–1045. doi:10.1007/s40879-020-00447-4.

- [12] D. Huybrechts, *Lectures on K3 Surfaces*, Cambridge Studies in Advanced Mathematics 158, Cambridge University Press, 2016. doi:10.1017/CBO9781316594193.
- [13] C. Ingalls and A. Kuznetsov, On nodal Enriques surfaces and quartic double solids, *Math. Ann.* **361** (2015), 107–133. doi:10.1007/s00208-014-1066-y.
- [14] J. Kollár, *Rational Curves on Algebraic Varieties*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 32, Springer, 1996, Chapter IV, §5.
- [15] N. Manolache, Cohen–Macaulay nilpotent schemes, arXiv:math/0312514 (2003).
- [16] G. Martin, G. Mezzedimi, and D. C. Veniani, Nodal Enriques surfaces are Reye congruences, *J. Reine Angew. Math.* **808** (2024), 49–65. doi:10.1515/crelle-2023-0092.
- [17] W. Fulton, *Intersection Theory*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 2, Springer, 1998, §5.1.
- [18] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer, 1991.
- [19] The Stacks Project Authors, *The Stacks Project*, Section 31.20, “The normal cone of an immersion,” Tag 062Z, <https://stacks.math.columbia.edu/tag/062Z>.
- [20] The Stacks Project Authors, *The Stacks Project*, Section 38.21, “Flattening stratifications,” Tag 052F, <https://stacks.math.columbia.edu/tag/052F>.